Lecture Notes: Introduction to Astrophysics and Cosmology

Based on lectures by **Dr. Shmuel Bialy** in 2025–26
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These notes were typed in the 2026 winter semester *Introduction to Astrophysics and Cosmology* course taught by**Dr. Shmuel Bialy** at the Technion.

The lectures were given in Hebrew and were live-translated by me to English. The document is provided as is and likely contains many errors.

If you find any mistakes or typos please let me know at `danielbreger@campus.technion.ac.il`.

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Lecture 1.

Thu, October 30, 2025

1 introduction

Since astrophysics is a very wide subject we won't be able to dive deep on each subject. The course is introductory so we'll be skipping some things like full solutions to integrals and the such.

1.1 So what is astrophysics?

Astrophysics and Cosmology could be broken down into:

Physics of stars

Solar systems planets, comets...

ISM Inter-Stellar Medium, Inter-Galactic Medium

Cosmology development, the big bang, dark energy/matter

Exotic objects black holes, neutron stars, white dwarfs...

Relativity gravitational waves, gravitational lensing

Galaxies types, development, active galactic nuclei, quasars

Explosions novas, supernovas, gamma ray bursts

and the methods used to research those things include

Observations
- **Telescopes** IR, radio, xray, gamma rays...
- **Gravitational waves**
- **Neutrinos**
- **Other particles (via probes)**

Labs

Computer simulations hydrodynamics, N-body simulations...

Analytical theory

We'll focus mostly on the analytical theory, but we'll mention the others too.

Lecture 2.

Mon, November 3rd, 2025

1.2 Regular (Baryonic) Matter

In our normal day-to-day life we meet protons, neutrons, and electrons. There are other particles we come across, such as neutrinos, muons, taus, etc. Another particle we meet often is the photon, and some others we won't get into.

When the universe came into being in the big bang quarks and photons were created. Those quarks then combined into protons and neutrons. Protons are basically hydrogen nuclei, so hydrogen was created in the big bang. For a short period of time after the big bang there were conditions for nuclear fusion across the early universe, so about 10% of the matter created from the big bang ended up being helium, and in terms of mass it was about 25% (Because Helium is heavier than hydrogen). Elements heavier than helium weren't created at any significant quantities in the big bang. If we plot this on a graph:

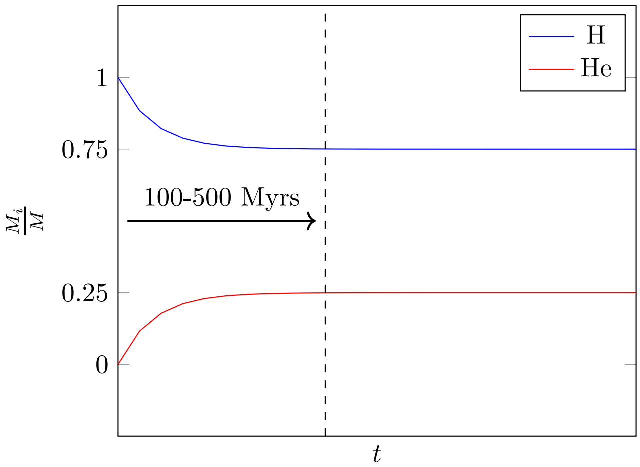


Figure 2.1: The portion of mass of the universe by element in the universe's early times.

The portion of mass of hydrogen of the total mass is denoted by X, hydrogen by Y, and metals, which are all the elements heavier than helium and are denoted by $Z = \frac{M_{metals}}{M}$. For our sun:

$$X_\odot = 0.7381 \tag{2.1}$$
$$Y_\odot = 0.2485 \tag{2.2}$$
$$Z_\odot = 0.0134 \tag{2.3}$$

The above was a discussion for gas and plasma. There is another kind of matter in space called dust. Dust is similar to the dust we know. It's particles about a micrometer in diameter made of water, carbon, silica, etc. Why do we mention this? In terms of the mass of the metals in the universe, about half of it is dust. Moreover, dust is a significant factor in observations of the universe, either in interfering in our ability to see things or in creating conditions that allow us to infer information. Dust also aids the creation of molecules by speeding up processes which are very slow in gasses.

An important definition for us will be the 'average mass' which we'll use in many formulas. Assume we have a box filled with hydrogen and some helium and some metals, all with the relative quantities that are in the sun. What would the average mass per particle be? Approximately a proton mass. And if the gas in the box is ionized? 0.5 proton masses. This is because we doubled the number of particles by freeing the electrons, which have approximately 0 mass relatively to the protons. This was a back of the napkin approximation just for sanity check and to have an idea of what we're expecting. The exact calculations:

$$\overline{m} = \frac{\sum\limits_j N_j m_j}{\sum\limits_j N_j} \tag{2.4}$$

Where j is each element type, N is their number, and m their mass. Rewriting in terms of X, Y, Z:

$$\frac{1}{\overline{m}} = \frac{\sum N_j}{\sum N_j m_j} \tag{2.5}$$

$$= \frac{M_j}{M_{tot}} \cdot \sum \frac{N_j}{M_j} \tag{2.6}$$

$$= \frac{1}{m_H}\left(X + \frac{1}{4}Y + \frac{1}{A_Z}Z\right) \tag{2.7}$$

where

$$M_j = \begin{cases} N_H m_H & H \\ N_{He} \cdot 4m_H & He \\ N_Z A_Z & else \end{cases} \tag{2.8}$$

Sometimes the following definition is used:

$$\mu = \frac{\overline{m}}{m_H} \tag{2.9}$$

and if we plug in the numbers for the sun we indeed find that it's very close to 1. We won't do this explicitly because there's a correction for the formula for a non-neutral gas, which our sun is not. For a completely ionized gas, i.e all the electrons are free, we just need to change N_j:

$$N_j = \begin{cases} 1 \to 2 & H \\ 1 \to 3 & He \\ 1 \to \left\langle \frac{A_Z}{2} \right\rangle & Else \end{cases} \tag{2.10}$$

Which gives us, for a completely ionized gas:

$$\mu = \left[2X + \frac{3}{4}Y + \frac{1}{2}Z\right]^{-1} \tag{2.11}$$

and if you insert the numbers for our sun we find

$$\langle A \rangle = 15 - 16 \tag{2.12}$$

$$\mu_{\odot} \simeq 0.6 \tag{2.13}$$

Lecture 3.

Thu, November 6th, 2025

1.3 The meaning of the symbol ≃

The symbol ≃ usually implies a good understanding of the physics behind the calculations. It says there's a good understanding of what can be considered negligible and what can't. The luminosity of the sun follows

$$L_\odot = \dot{E}_{\text{nuc}} + |\dot{E}_G| \tag{3.1}$$

Observationally we can measure the luminosity and find

$$L_\odot = 3.8 \times 10^{33}\,\text{erg}\,\text{s}^{-1} \tag{3.2}$$

and we can say that

$$L_\odot \simeq \dot{E}_{\text{nuc}} \tag{3.3}$$

Let's look at the total energy the Sun will output in its entire life E_{tot}.

$$\begin{aligned} E_{tot} &= L_\odot \cdot t_\odot & (3.4)\\ &\simeq (3.8 \times 10^{33}\,\text{erg}\,\text{s}^{-1})(4.6\,\text{Gyr}) & (3.5)\\ &= 5.6 \times 10^{50}\,\text{erg} & (3.6) \end{aligned}$$

Let's look at what contributes more to the energy of the sun, the gravitational or the nuclear energy.

$$\begin{aligned} |E_G| &= \int \frac{GM(r)}{r}\,\mathrm{d}m & (3.7)\\ &= \int 4\pi\rho GM(r) r\,\mathrm{d}r & (3.8)\\ (\text{assuming } \rho = \text{const.}) &= \frac{3}{5} G \frac{M_\odot^2}{R_\odot} & (3.9)\\ &= 2 \times 10^{48}\,\text{erg} & (3.10) \end{aligned}$$

To estimate the nuclear energy the sun produces we can look at the nuclear fusion process in which $4H \to He$ which gives a mass difference of

$$\begin{aligned} \Delta m &= M(4H) - M(He) & (3.11)\\ \to \Delta E &= \Delta m c^2 & (3.12) \end{aligned}$$

defining

$$\eta = \frac{\Delta E}{M_{He} c^2} = 0.7\% \tag{3.13}$$

then the nuclear energy produced in the sun is

$$\eta \cdot M_\odot c^2 \tag{3.14}$$

and using this we can find the age of the sun

$$t_\odot = \frac{E_{nuc}}{L_\odot} \approx 100\,\text{Gyr} \tag{3.15}$$

but since the sun hasn't burned through all the hydrogen it has access to, we can replace

$$M_\odot \to M_{core} \approx 5\,\% M_\odot \tag{3.16}$$

which would give us a better estimate.

2 The Structure of Stars

We'll use the Sun as an example, then we'll hit equations.

2.1 The Sun

Our goal in the coming weeks will be to derive differential equations to figure out how the different profiles behave. As for some numbers:

$$R_\odot = 7\times10^{10}\,\mathrm{cm} \qquad T_C = 1.5\times10^{7}\,\mathrm{K}$$
$$\rho_C = 150\,\mathrm{g\,cm^{-3}} \qquad \overline{\rho} = 1.4\,\mathrm{g\,cm^{-3}}$$
$$P_C = 10^{11}\,\mathrm{Atm} \qquad M_\odot = 2\times10^{33}\,\mathrm{g}$$
$$L_\odot = 3.8\times10^{33} erg/s$$

2.2 Star Structure Equations

The equations describing the structure of stars are:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -G\frac{M_r\rho}{r^2} \tag{3.17}$$
$$\frac{\mathrm{d}M_r}{\mathrm{d}r} = 4\pi r^2\rho \tag{3.18}$$
$$\frac{\mathrm{d}L_r}{\mathrm{d}r} = 4\pi r^2\rho\epsilon \tag{3.19}$$
$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3\kappa\rho}{4acT^3}\frac{L_r}{4\pi r^2} \tag{3.20}$$

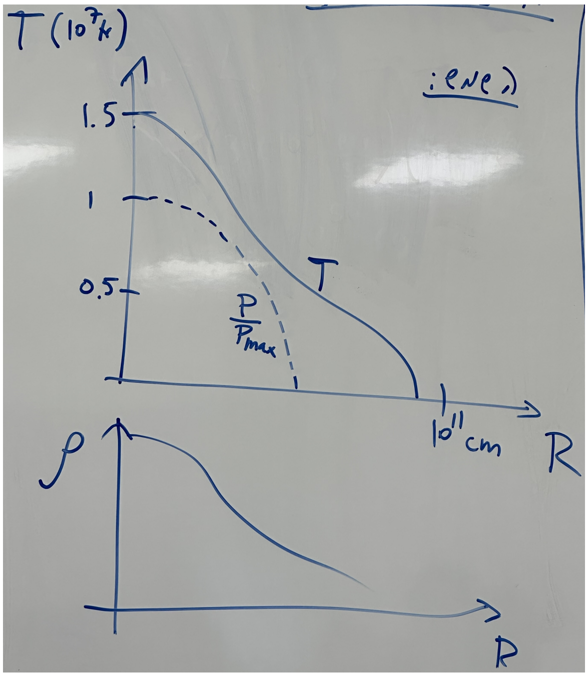


Figure 3.1: The temperature in the sun as a function of the distance from it core, the relative pressure compared to the core, and the density.

The first equation is called the hydrostatic equation and is equivalent to conservation of momentum, the second is the continuum equation and is equivalent to conservation of mass, the third is the energy production equation and is equivalent to conservation of energy, and the final one is the radiation diffusion equation.

2.3 The Hydrostatic Equation

The equation can be derived from an equality of forces. We look at a shell at radius r with mass $\mathrm{d}m$ and ask which forces act on it. On one hand there's gravitation pulling it inwards and on the other side there's a force resulting from a difference in pressure.

$$|\mathrm{d}F_g| = \frac{GM_r\,\mathrm{d}m}{r^2} \tag{3.21}$$
$$= \frac{GM_r}{r^2}4\pi r^2\rho\,\mathrm{d}r \tag{3.22}$$
$$|\mathrm{d}F_P| = A\cdot(P(r)-P(r+\mathrm{d}r)) \tag{3.23}$$
$$= 4\pi r^2\left(-\frac{\mathrm{d}P}{\mathrm{d}r}\,\mathrm{d}r\right) \tag{3.24}$$

equating the two forces we can find

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho\frac{GM_r}{r^2} \tag{3.25}$$

which is indeed the hydrostatic equation.

Applying The Equation To Earth

As a back of the envelope calculation, we can use the equation to find the gravitational acceleration of earth. Noticing that

$$\frac{GM_r}{r^2} = g \tag{3.26}$$

we can rewrite the hydrostatic equation as

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\rho g \tag{3.27}$$

We can also find the speed of sound

$$c_s = \sqrt{\gamma \frac{\mathrm{d}P}{\mathrm{d}\rho}} \tag{3.28}$$

which means

$$\frac{\mathrm{d}P}{\mathrm{d}\rho} \approx c_s^2 \tag{3.29}$$

so

$$\begin{aligned} g &\sim \frac{\mathrm{d}P}{\mathrm{d}\rho}\frac{1}{h} && (3.30)\\ &= \frac{\left(340\,\mathrm{m\,s^{-1}}\right)^2}{10\,\mathrm{km}} && (3.31)\\ &= 10\,\mathrm{m/s}^2 && (3.32) \end{aligned}$$

Lecture 4.

Mon, November 10th, 2025

2.4 The State Equation

The gas in stars is to a good approximation an ideal gas so we can use the state equation of an ideal gas

$$P = Nk_BT \tag{4.1}$$

we can multiply and divide by the average mass of each particle

$$P = \frac{n\overline{m}}{\overline{m}}k_BT = \frac{\rho}{m_p\mu}k_BT \tag{4.2}$$

A quick aside on why the state equation is true

This is taken from Feynman's lectures on physics volume 2 chapter 39–2. Assume we have a wall of area A and particles with density n, each having velocity v_x and they impact the wall head on. As we know $F = \frac{\mathrm{d}p}{\mathrm{d}t}$. The distance the particles travel over a period of time of δt is $\delta x = v_x\delta t$. So the rate at which particles hit over the same time period is $\delta N = \frac{1}{2}nAv\delta t$. Each particles transfers a momentum of $2p_1 = 2mv$ therefore the total momentum transferred is

$$\delta p = mnAv_x^2\delta t \tag{4.3}$$

therefore the force is

$$F = \frac{\mathrm{d}p}{\mathrm{d}t} = mnAv_x^2 \tag{4.4}$$

and the pressure is

$$P = \frac{F}{A} = mnv_x^2 \tag{4.5}$$

and if the velocity isn't a constant but rather there's a distribution of velocities then

$$P = mn\langle v_x^2\rangle \tag{4.6}$$

when in 3 dimensions

$$\langle v^2\rangle = \langle v_x^2\rangle + \langle v_y^2\rangle + \langle v_z^2\rangle = 3\langle v_x^2\rangle \tag{4.7}$$

So the pressure is

$$P = \frac{1}{3}mn\langle v^2\rangle \tag{4.8}$$

If we define temperature as follows

$$\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}k_BT \tag{4.9}$$

and then insert this into our equation for pressure we get

$$P = nk_BT \tag{4.10}$$

Thermal energy is

$$U = \frac{1}{2}m\langle v^2\rangle N \tag{4.11}$$

so its density is

$$u = \frac{1}{2}m\langle v^2c\rangle n \tag{4.12}$$

therefore we can insert this into the expression for pressure we have and find

$$P = \frac{2}{3}u \tag{4.13}$$

This is correct for mono-atomic gas such as hydrogen. In the more complicated case we can define the adiabatic index

$$\gamma_{ad} = \frac{C_p}{C_v} = 1 + \frac{2}{f} \tag{4.14}$$

where f is the number of degrees of freedom, and then we can find that

$$P = (\gamma_{ad} - 1)u \tag{4.15}$$

2.5 Generalizing To Radiation Pressure

The distribution of the velocities of the massive particles is the Boltzmann distribution. On the other hand, the distribution of the momenta of the photons is the Planck distribution:

$$\frac{\mathrm{d}n}{\mathrm{d}\nu} = n_\nu = \frac{8\pi\nu^2}{c^3}\frac{1}{e^{\frac{h\nu}{k_BT}} - 1} \tag{4.16}$$

and

$$u_\nu = n_\nu h\nu = \frac{8\pi h\nu^3}{c^3}\frac{1}{e^{\frac{h\nu}{k_BT}} - 1} \tag{4.17}$$

To get the pressure, rather than doing an entirely new development, we can say our previous development of the pressure using a wall and particles was good it just needs some adjustments. We had:

$$P = \frac{1}{3}mn\langle v^2\rangle = \langle\frac{1}{3}nv \cdot mv\rangle = \langle\frac{1}{3}nv \cdot p\rangle \tag{4.18}$$

inserting the momentum and velocity of photons we find

$$P = \langle\frac{1}{3}nc\frac{h\nu}{c}\rangle \tag{4.19}$$

$$= \langle\frac{1}{3}nh\nu\rangle \tag{4.20}$$

$$= \frac{1}{3}\langle u\rangle \tag{4.21}$$

$$= \frac{1}{3}u \tag{4.22}$$

And if we equate this to the more general pressure equation using the adiabatic constant we find

$$\gamma_{\text{photons}} = \frac{4}{3} \tag{4.23}$$

to rewrite this in terms of temperature we remember that

$$u = \int_0^\infty u_\nu \, \mathrm{d}\nu = \int \frac{8\pi h\nu^3}{c^3}\frac{1}{e^{\frac{h\nu}{k_BT}} - 1} \tag{4.24}$$

$$= [x = \frac{h\nu}{k_BT}] = 8\pi\frac{(kT)^4}{(hc)^3}\int_0^\infty x^3 \frac{\mathrm{d}x}{e^x - 1} \tag{4.25}$$

$$= 8\pi\frac{(kT)^4}{(hc)^3} \cdot \frac{\pi^4}{15} \tag{4.26}$$

we can define a constant

$$a = \frac{8\pi^5 k_B^4}{15\,(hc)^3} \tag{4.27}$$

and have

$$u = aT^4 \tag{4.28}$$

and find that the total pressure is

$$P_{tot} = nk_BT + \frac{1}{3}aT^4 \tag{4.29}$$

2.6 Nuclear Fusion

quick look at the periodic table. quick look at the standard model. the dominant nuclear fusion process is the pp-chain. The efficiency of the process is

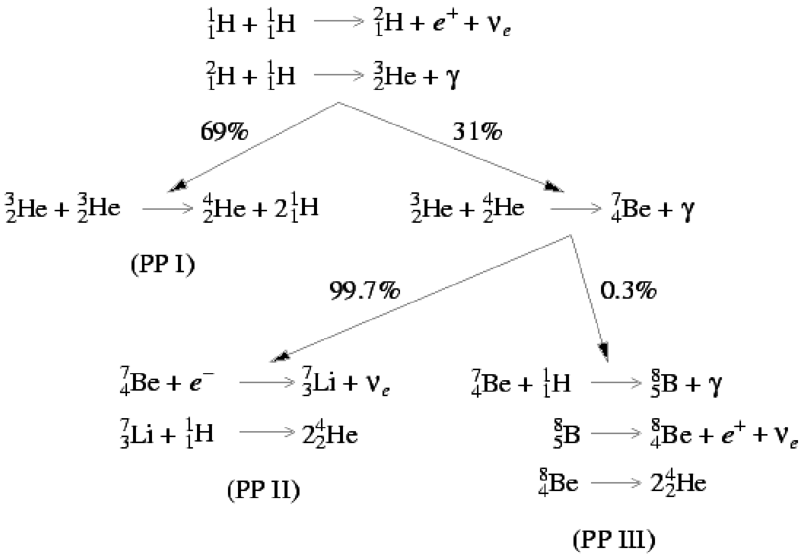


Figure 4.1: https://burro.cwru.edu/academics/Astr221/StarPhys/ppchain.html

$$\eta = \frac{E}{Mc^2} = \frac{26.7MeV}{4m_pc^2} = 0.7\% \tag{4.30}$$

for context of how much energy that is, we know that the sun's luminosity is $l_\odot = 4 \times 10^{33}$ erg/sec. This means that per second it produces 4×10^{26} J, which is 10^6 times the yearly consumption of all humanity.

Lecture 5.

Thu, November 13th, 2025

To find the luminosity we need to also figure out the pace of the processes, which we'll call r_{xi} whose units are $\mathrm{s}^{-1}\mathrm{cm}^{-3}$, x indicating the target particle being hit by the particle i. First though let's check if nuclear fusion can happen in the sun at all. The electrostatic potential (in cgs) is given by

$$U = \frac{e^2}{r} = \frac{1.44\,\mathrm{MeV}}{(r/fm)} \tag{5.1}$$

where

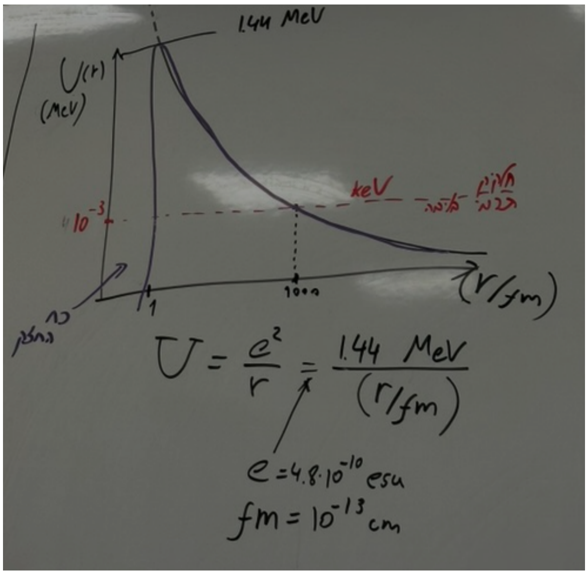
$$e = 4.8 \times 10^{-10}\,\mathrm{esu} \qquad\qquad fm = 10^{-13}\,\mathrm{m} \tag{5.2}$$

and the temperature at the core of the sun is $T_c \simeq 1 \times 10^7\,\mathrm{K}$ and using the expression for the energy at the core

$$\frac{3}{2}k_B T = E \implies E_c \sim keV \tag{5.3}$$

This means that the thermal particles have only 0.1% of the energy required to jump the energy barrier imposed by the coulomb force's repulsion to reach the area the strong force attracts them. Seemingly if the system were purely classical, fusion could not happen. Let's consider an edge case where maybe fusion happens because a random small number of particles just happen to have a high enough energy to pass the barrier. To do that we can calculate how many particles on average there are that have an energy with approximately a mega-electronvolt of energy. For a rough estimate we can use statistical mechanics

$$e^{-\frac{E}{k_B T}} \sim e^{-1000} = 10^{-435} \tag{5.4}$$



where we inserted the temperature at the core of the sun and the energy we want to have to breach the barrier. The number is so small that there would on average be less than one particle in the entire universe that would have that energy. For that reason we didn't calculate it accurately, since it wouldn't have mattered. We'll need to take into consideration the effects of quantum tunneling, but before we do that, we need to discuss the action cross-section. We'll imagine a ball x with effective surface area σ being bombarded with particles i.

The number of particles hitting the area is given by

$$\mathrm{d}N = \sigma v \delta t \cdot n_i \tag{5.5}$$

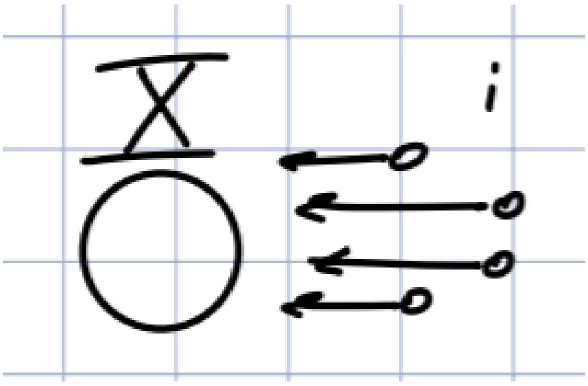
so the rate at which they hit is

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \sigma v n_i \tag{5.6}$$

If we have an environment with many x particles then

$$r_{xi} = \sigma v n_i \cdot n_x \tag{5.7}$$

where n_x is the density of the particles x. In the more general case if we have a distribution of velocities then we have



$$v \implies v(E) \qquad\qquad n_i \implies \frac{\mathrm{d}n_i}{\mathrm{d}E}\,\mathrm{d}E \tag{5.8}$$

and then

$$r_{xi} = \int \sigma(E) v(E) n_x \frac{\mathrm{d}n_i}{\mathrm{d}E}\,\mathrm{d}E \tag{5.9}$$

with the Stefan-Boltzmann law being

$$\frac{\mathrm{d}n_i}{\mathrm{d}E} = \frac{2n_i}{\sqrt{\pi}} \frac{\sqrt{E}}{(k_B T)^{3/2}} e^{-\frac{E}{k_B T}} \tag{5.10}$$

our goal now is to find $\sigma(E)$. For quantum tunneling we'll write

$$\sigma = \sigma_0 P \tag{5.11}$$

where P is the chance to tunnel and σ_0 is the effective area. As for σ_0:

$$\sigma_0 \approx \pi R^2 \tag{5.12}$$
$$= \pi \lambda_{DB}^2 \tag{5.13}$$
$$= \pi \left(\frac{h}{p}\right)^2 \tag{5.14}$$
$$= \pi \frac{h^2}{2\mu E} \tag{5.15}$$

where μ is the reduced mass, since we're working with a collision of two particles. or, in a general more accurate form

$$\sigma_0 = \frac{S}{E} \tag{5.16}$$

To find the chance to tunnel we'll refer to quantum mechanics but this isn't a course on quantum mechanics so it's just a quick overview.

The Schrödinger equation is

$$\frac{\hbar^2}{2\mu}\nabla^2\Psi = (V(r) - E)\,\Psi \tag{5.17}$$

where our energy is

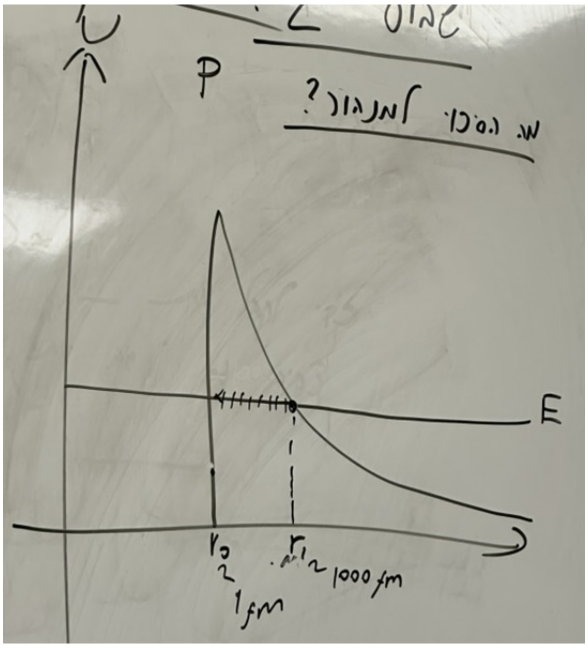
$$E = \frac{3}{2}k_B T = \frac{z_1 z_2 e^2}{r_1} \tag{5.18}$$

and our potential is

$$V(r) = \begin{cases} \frac{Z_A Z_B e^2}{r} & r > r_0 \\ 0 & r < r_0 \end{cases} \tag{5.19}$$

Solving for this potential is complicated. We'll approximate it using a step function. Its height should be the average of the potential

$$\langle V(r)\rangle = \frac{\int_{r_0}^{r_1} 4\pi r^2 V(r)\,\mathrm{d}r}{\int_{r_0}^{r_1} 4\pi r^2\,\mathrm{d}r} = \frac{3}{2}E \tag{5.20}$$



and a step function is a problem we know how to solve. Inside the step we have

$$\frac{\hbar^2}{2\mu}\nabla^2\Psi = \left(\frac{3}{2}E - E\right)\Psi \tag{5.21}$$

and since we're working in spherical coordinates

$$\frac{\hbar}{2\mu r}\frac{\partial^2}{\partial r^2}(r\Psi) = \frac{1}{2}E\Psi \tag{5.22}$$

whose solution is

$$\Psi = A\frac{e^{\beta r}}{r} \qquad \beta = \frac{\sqrt{\mu E}}{\hbar} \tag{5.23}$$

and therefore the chance to find a particle in an area δr is proportional to $|\Psi|\,\delta r$ and the probability of tunneling is

$$P_{\text{tunnel}} = \frac{4\pi r_0^2 \delta r\,|\Psi(r_0)|^2}{4\pi r_1^2 \delta r\,|\Psi(r_1)|^2} = \frac{e^{2\beta r_0}}{e^{2\beta r_1}} = e^{2\beta(r_0 - r_1)} \approx e^{-2\beta r_1} \tag{5.24}$$

inserting the parameters we found earlier we find

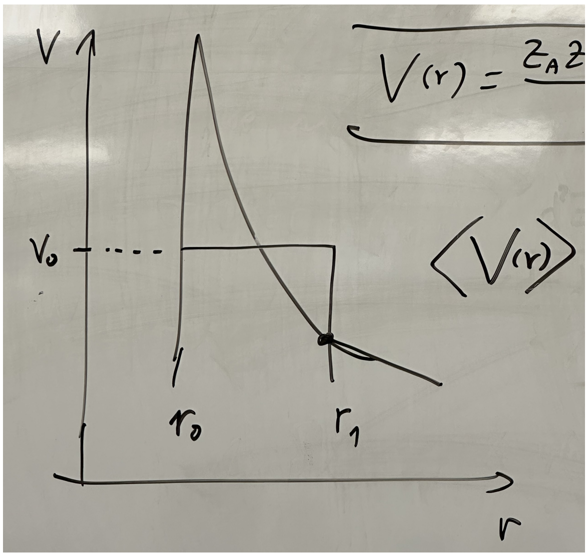
$$P \approx e^{-\frac{2\sqrt{\mu E}}{\hbar}\frac{Z_A Z_B e^2}{E}} \tag{5.25}$$

while the exact solution is

$$P = e^{-\sqrt{E_G/E}} \tag{5.26}$$

where

$$E_G = (\pi\alpha Z_A Z_B)^2\, 2\mu c^2 \qquad \alpha = \frac{e^2}{\hbar c} \approx \frac{1}{137} \tag{5.27}$$


V
V_0
r_0
r_1
r
$V(r) = Z_A Z$
$\langle V(r) \rangle$

Lecture 6.

Mon, November 17th, 2025

It's worth noting that in the discussion on the probability of tunneling it is more likely the higher the energy, but the cross section for fusion becomes smaller, so it's a tradeoff. Specifically for the pp-chain the numbers are

$$Z_A = Z_B = 1 \qquad \mu = \frac{1}{2}m_p \qquad \mu c^2 = 0.94\,\mathrm{GeV} \tag{6.1}$$

giving us

$$E_G \approx 500\,\mathrm{keV} \tag{6.2}$$

therefore the probability of tunneling is

$$P = e^{-\sqrt{500}} = 2 \times 10^{-10} \tag{6.3}$$

Let's check what this means for the sun. The number of particles in the sun's core is

$$N = 0.1\frac{M_\odot}{m_p} \sim 10^{48} \tag{6.4}$$

So even though the probability of tunneling is low, it definitely happens. To find the rate of the interactions we insert this into 5.9:

$$r_{xi} = \int \sigma n \frac{\mathrm{d}n}{\mathrm{d}v}\,\mathrm{d}v \tag{6.5}$$

$$= \int \sigma \sqrt{\frac{2E}{\mu}} \frac{\mathrm{d}n}{\mathrm{d}E}\,\mathrm{d}E \tag{6.6}$$

$$\tag{6.7}$$

where

$$\frac{\mathrm{d}n}{\mathrm{d}E} = n_i \frac{2}{\sqrt{\pi}} \frac{\sqrt{E}}{(k_B T)^{3/2}} e^{-\frac{E}{k_B T}} \tag{6.8}$$

$$\sigma = \frac{S}{E} e^{-\sqrt{E_G/E}} \tag{6.9}$$

$$S = \frac{\pi \hbar^2}{2\mu} \tag{6.10}$$

giving us

$$r_{xi} = n_i n_x \left(\frac{1}{k_B T}\right)^{3/2} \sqrt{\frac{8}{\pi\mu}} \int e^{-\left(E/K_B T + \sqrt{E_G/E}\right)}\,\mathrm{d}E \tag{6.11}$$

to figure out which particles are most responsible for the fusion process, the high energy ones or the low energy ones, if we designate

$$f(x) = e^{-\left(E/K_B T + \sqrt{E_G/E}\right)} \tag{6.12}$$

then it looks like and the peak is found at

$$E_0 = \left(\frac{k_B T}{2}\right)^{2/3} E_G^{1/3} \tag{6.13}$$

and specifically for the Sun

$$E_0 = 5\,\mathrm{keV} \tag{6.14}$$

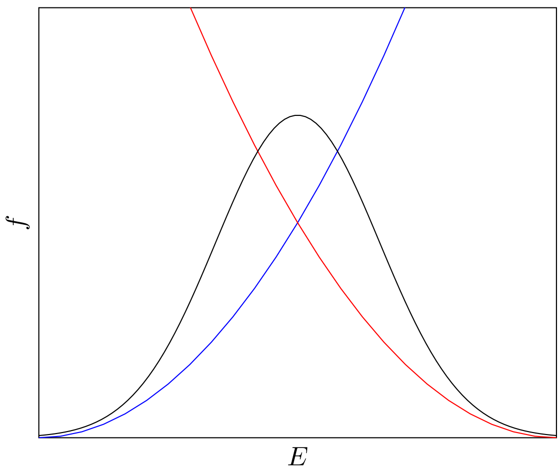


Figure 6.1: The portion of mass of the universe by element in the universe's early times.

Next we want to solve 6.11. It's not analytically solvable so we approximate it. What's often done in astrophysics is that we approximate things to a power law. Our function looks like this:

$$r_{xi} = An^2 f(T) \tag{6.15}$$

a power law approximation for this would look like

$$r_{xi} \approx An^2 T^\beta \tag{6.16}$$

Why do we do this? Often in astrophysics we look at relationships between values of parameters which span very large ranges, often entire orders of magnitude. for this reason instead of plotting regular linear graphs, we use log-log graphs. If we take a taylor series at some point and look at the first order it's a linear function. A linear line on a log-log graph fits a function of the form T^β. Back to the case of our Sun

$$r_{xi} = An_i n_x f(T) = A\rho_i \rho_x T^\beta \tag{6.17}$$

and if we define

$$x_i = \frac{\rho_i}{\rho} \qquad x_x = \frac{\rho_x}{\rho} \tag{6.18}$$

then

$$r_{xi} = r_0 x_i x_x \left(\frac{\rho}{\rho_0}\right)^2 \left(\frac{T}{T_0}\right)^\beta \tag{6.19}$$

where r_0, ρ_o, T_0 are parameters we find from the fit. The amount of energy produced from all the fusion is

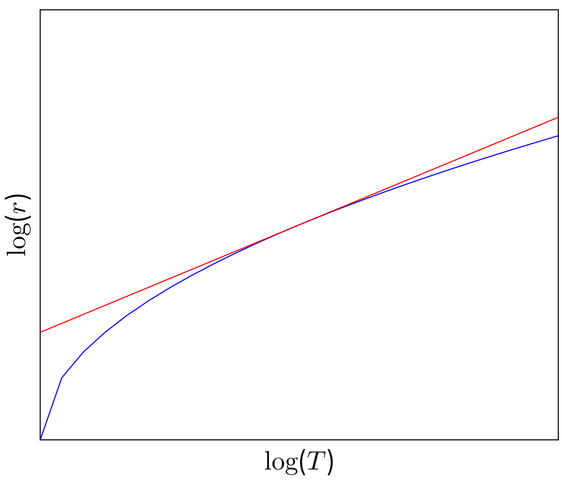


Figure 6.2: The portion of mass of the universe by element in the universe's early times.

$$\mathrm{d}E = E_0 r_{xi}\,\mathrm{d}t\,\mathrm{d}V \tag{6.20}$$

and inserting what we found so far

$$\mathrm{d}E = E_0 r_0 x_i x_x \left(\frac{\rho}{\rho_0}\right)^2 \left(\frac{T}{T_0}\right)^\beta \mathrm{d}V\,\mathrm{d}t \tag{6.21}$$

which we are able to rewrite to an integral over the mass by using $\mathrm{d}m = \rho\,\mathrm{d}V$:

$$\mathrm{d}E = \left(\frac{E_0 r_0}{\rho_0}\right) x_i x_x \left(\frac{\rho}{\rho_0}\right) \left(\frac{T}{T_0}\right)^\beta \mathrm{d}m\,\mathrm{d}t \tag{6.22}$$

where we can define

$$\epsilon_0 = \frac{E_0 r_0}{\rho_0} \tag{6.23}$$

which has units of energy produced per gram material per second, giving us

$$\frac{\mathrm{d}^2 E}{\mathrm{d}m^2} t = \epsilon_0 x_i x_x \frac{\rho}{\rho_0} \left(\frac{T}{T_0}\right)^\beta \tag{6.24}$$

and if we insert $\beta = 4$ which fits our Sun and

$$\epsilon_0 = 0.1\,\mathrm{erg\,g^{-1}\,s^{-1}} \qquad \rho_0 = \mathrm{g/cm^3} \qquad T_0 = 10^7\,\mathrm{K} \qquad x_i = x_x = 0.7 \tag{6.25}$$

then we find

$$\epsilon_\odot = 7.4\,\mathrm{erg\,g^{-1}\,s^{-1}} \tag{6.26}$$

then the luminosity produced is

$$\mathrm{d}L = \epsilon\,\mathrm{d}m = \epsilon \cdot 4\pi r^2\,\mathrm{d}r\rho \tag{6.27}$$

which gives us

$$\frac{\mathrm{d}L}{\mathrm{d}r} = \epsilon \cdot 4\pi r^2 \rho \tag{6.28}$$

2.7 Diffusion of Photons

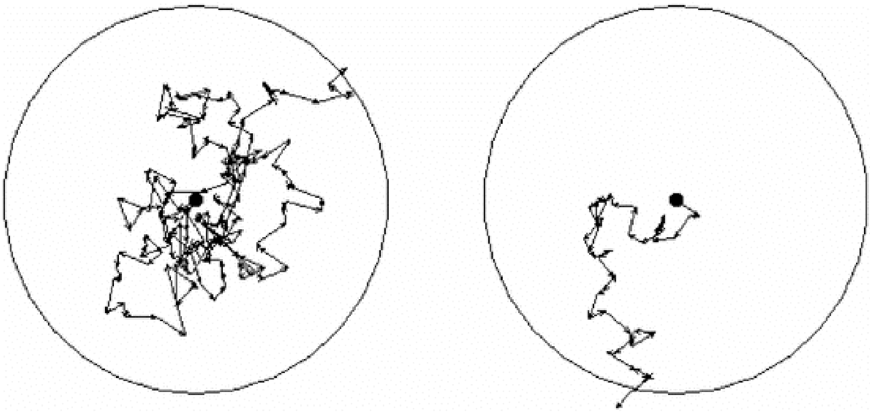


Figure 6.3: Example of Compton scattering Brownian motion (Random walk) of a photon.

Suppose we have a star and inside a photon is created in the middle and moves around inside until it reaches the outside. This process is called a random walk. Each step is of length λ and the star's radius is R.

$$R^2 = |\mathbf{R}|^2 = \left|\sum \mathbf{r}_i\right|^2 \tag{6.29}$$

$$= \sum_i \sum_j \mathbf{r}_i \cdot \mathbf{r}_j \tag{6.30}$$

$$= \sum_{i=1}^N |r_i|^2 + \sum_{i \neq j} \mathbf{r}_i \cdot \mathbf{r}_j \tag{6.31}$$

$$= N\lambda^2 \tag{6.32}$$

therefore the number of steps until the photon escapes is

$$N = \left(\frac{R}{\lambda}\right)^2 \tag{6.33}$$

some typical numbers are

$$R = R_{\odot} \qquad\qquad \lambda \approx 0.1\,\mathrm{cm} \tag{6.34}$$

which gives us

$$N \approx 5 \times 10^{23} \tag{6.35}$$

if we want the distance the photon will travel in total then

$$d = \lambda \cdot N \approx 4 \times 10^{22}\,\mathrm{cm} = 16\,\mathrm{kpc} \tag{6.36}$$

and the time it takes the photon to travel this distance is

$$t = d/c \approx 50\,000\,\mathrm{yr} \tag{6.37}$$

If want a more accurate description of the process, we can use the diffusion equation. We won't develop it from scratch but begin from Fick's Diffusion Law (A development from scratch can be found in Feynman's lectures volume 1 chapter 42):

$$\mathbf{J} == -D\nabla n \tag{6.38}$$

where J is the flux of particles, $D = \frac{\lambda v}{3}$ is the diffusion parameter. We'll use this formula for photons. The free traversal is given by $\lambda = \frac{1}{\sigma n} = \frac{1}{\kappa\rho}$ where σ is the action cross section and $\kappa = \frac{\sigma}{m}$ is the opacity. So for photons we'll perform the substitutions

$$v \to c \qquad D \to \frac{\lambda c}{3} = \frac{c}{3\kappa\rho} \qquad \rho \to u_{rad} = aT^4 \qquad J \to f_{rad} = \frac{L_r}{4\pi r^2} \tag{6.39}$$

which gives us

$$\frac{L_r}{4\pi r^2} = f_{rad} = -\frac{1}{3}\frac{c}{\kappa\rho}\frac{\mathrm{d}}{\mathrm{d}r}aT^4 \tag{6.40}$$

rearranging differentiating we find

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3}{4ac}\frac{\kappa\rho}{T^3}\frac{L_r}{4\pi r^2} \tag{6.41}$$

In conclusion we have four differential equations:

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -G\frac{M_r\rho}{r^2} \qquad\qquad \frac{\mathrm{d}M_r}{\mathrm{d}r} = 4\pi r^2\rho \tag{6.42}$$

$$\frac{\mathrm{d}L_r}{\mathrm{d}r} = 4\pi r^2\rho\epsilon \qquad\qquad \frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3}{4ac}\frac{\kappa\rho}{T^3}\frac{L_r}{4\pi r^2} \tag{6.43}$$

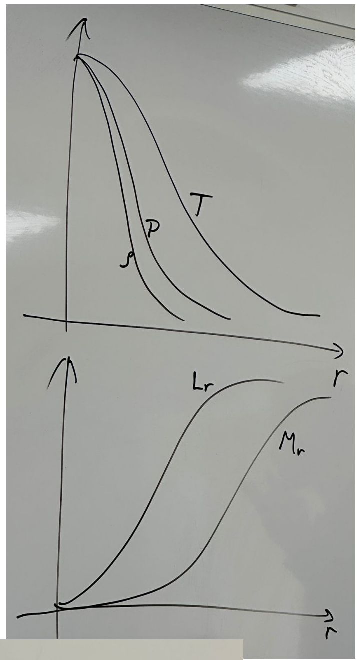
together with the state equation P, the energy production equation ϵ, and the opacity κ we have all the tools we need to describe the structure of stars.

Lecture 7.

Thu, November 20th, 2025

For an approximate set of solutions we can assume that

$$T, P, \rho\big|_{r=R_*} = 0 \qquad M_r = L_r\big|_{r=0} = 0 \qquad \begin{cases} M_r\big|_{R_*} = M \\ L_r\big|_{R_*} = L \end{cases} \tag{7.1}$$



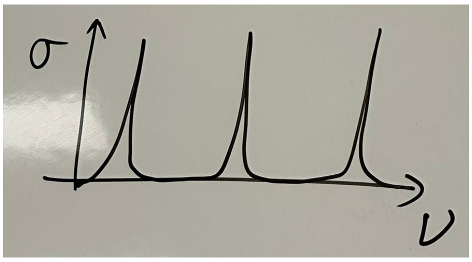
In the process of diffusion, photons can undergo several kinds of interactions

Bound-bound (Excitation) $X + \nu \to X^*$.

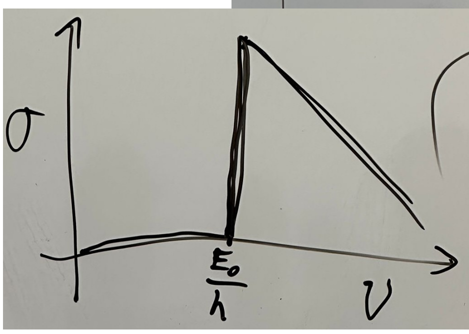
Bound-free (Ionization) $X^n + \nu \to X^{n+1} + e$.
With $\sigma = \sigma_0 \left(\frac{h\nu}{E_0}\right)^{-3}$ for hydrogen and atoms with one electons.

Free-free absorption (Reverse bremsstrahlung)

Free-free e-scattering (Thomson) which has a well known action cross section $\sigma_T = \frac{8}{3}\pi r_e^2$ where $r_e = \frac{e^2}{m_e c^2}$. Additionally, if there are ions in the area then if $\lambda \ll r_p$ there can happen Compton scattering and if $\lambda \gg r_p$there can happen Rayleigh scattering.



(a) Bound-bound (excitation)



(b) Bound-free (Ionization)

Lecture 8.

Mon, November 24th, 2025

3 L-M ties

Looking at the HR diagram we see that there appears to be a connection between mass and luminosity. We'll dwell on this for a while. This also connects to its life span and evolution. Observationally, if we plot the luminosity as a function of the mass, we find something that looks like this

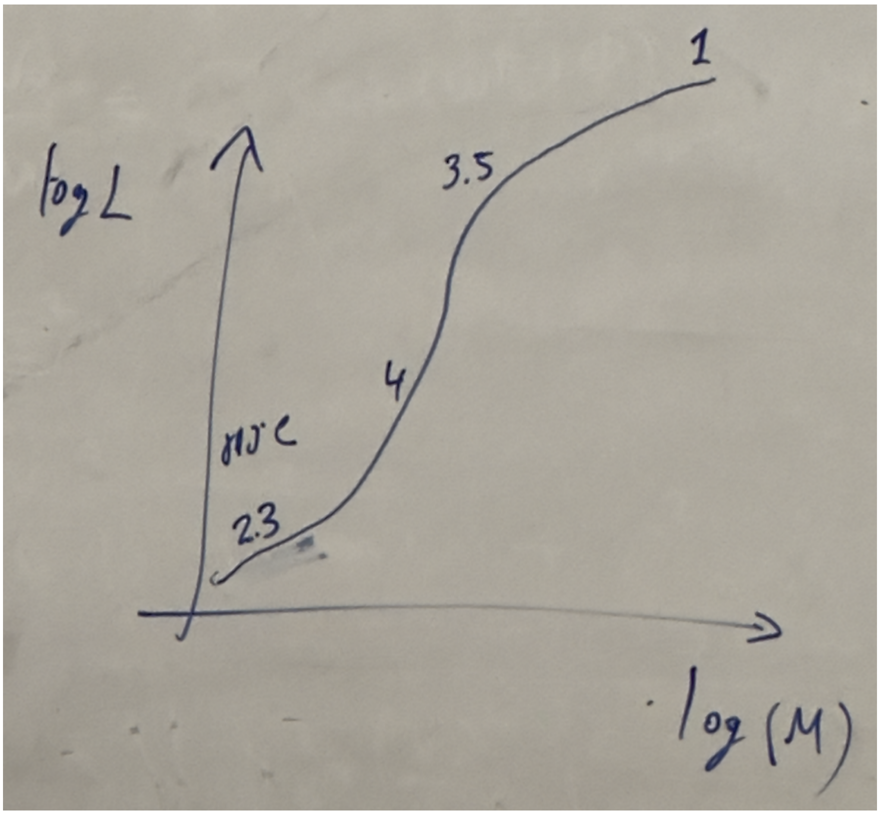


Figure 8.1: A diagram of the L-M relation

and using the relation

$$\frac{L}{L_\odot} = A\left(\frac{M}{M_\odot}\right)^\alpha \tag{8.1}$$

we can write this as a table To get a feeling for why this behaves like this, we'll do the following.

$M/M_\odot$	A	α
< 0.43	0.23	2.3
0.43–2	1	4
2–55	1.4	3.5
> 55	3.2×10^4	1

We'll assume

$$\kappa = \kappa_{\text{Thomson}} = \text{const.} \tag{8.2}$$

which is a good approximation for massive stars (2 to 55 solar masses). We'll also assume a power law of the form

$$M_r \propto r^{\alpha} \qquad P \propto r^{\beta} \qquad T \propto T^{r} \tag{8.3}$$

therefore the stellar structure equations become

$$\beta \frac{P}{r} \propto \frac{M\rho}{r^2} \qquad \alpha \frac{M}{r} \propto r^2 \rho \qquad \gamma \frac{T}{r} \propto \frac{L\kappa\rho}{r^2 T^3} \tag{8.4}$$

rearranging we find

$$P \propto \frac{M\rho}{r} \qquad M \propto r^3 \rho \qquad L \propto \frac{T^4 r}{\rho} \qquad P \propto \rho T \tag{8.5}$$

Inserting D into A we find

$$T \propto \frac{M}{r} \tag{8.6}$$

and inserting that into C

$$L \propto \frac{(M/r)^4 r}{\rho} = \frac{M^4/r^3}{\rho} \propto M^3 \tag{8.7}$$

For extremely massive stars ($> 55 M_{\odot}$), the radiation pressure is dominant, at which point D no longer holds and instead

$$P \propto T^4 \tag{8.8}$$

which inserted into A gives

$$T^4 \propto \frac{M\rho}{r} \tag{8.9}$$

and inserting that into C

$$L \propto \frac{(M\rho/r)\, r}{\rho} = M \tag{8.10}$$

3.1 The Most Massive Stars

Let's look at the outer layer of a very massive star. Using the radiation propagation equation

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{3}{4ac} \frac{\kappa\rho}{T^3} \frac{L_r}{4\pi r^2 c} \tag{8.11}$$

we can rearrange it to

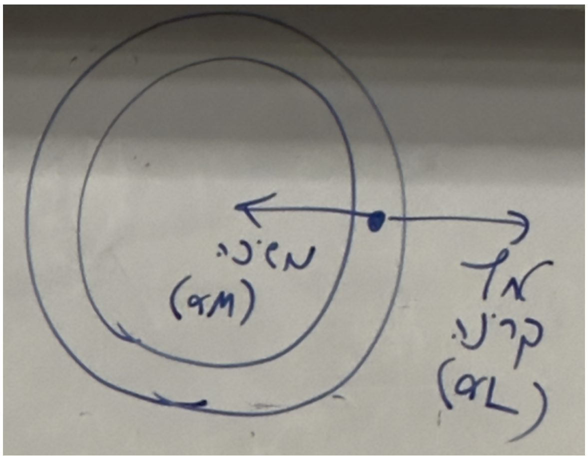
$$\frac{1}{3} \frac{\mathrm{d}}{\mathrm{d}r} \left(aT^4 \right) = -\kappa\rho \frac{L_r}{4\pi r^2 c} \tag{8.12}$$

and remembering that $\frac{1}{3} aT^4$ is the radiation pressure

$$\frac{\mathrm{d}P_{\mathrm{rad}}}{\mathrm{d}r} = -\kappa\rho \frac{L_r}{4\pi r^2 c} \tag{8.13}$$

on the other hand, from the hydrostatic equations we have

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{GM_r\rho}{r^2 c} \tag{8.14}$$



setting our two expression to equal we find

$$\frac{GM_r\rho}{r^2} = \kappa\rho \frac{L_r}{4\pi r^2 c} \tag{8.15}$$

isolating L_r and inserting $r = R$

$$L_r = 4\pi c \frac{GM}{\kappa} \tag{8.16}$$

this expression for luminosity is often called Eddington Luminosity. This expression gives us the maximum luminosity a star can have before it becomes unstable. If for a given star's parameters the observed luminosity is higher than the Eddington luminosity, then the radiation pressure will begin to overpower its hydrostatic pressure which will destabilize the star. Let's plug some numbers in to get a feel. Inserting the constants we have

$$L = 3.8 \times 10^4 \frac{M}{M_\odot} L_\odot \tag{8.17}$$

and for a mass of $M = 60M_\odot$

$$L = 2.3 \times 10^6 \,\mathrm{L}_\odot \tag{8.18}$$

3.2 The Lifespan Of Stars

For the Sun we can write

$$t_\odot \simeq \frac{\eta M_{\mathrm{core}} c^2}{L_\odot} \sim 10\,\mathrm{Gyr} \tag{8.19}$$

If we look at stars with different masses then

$$t_* \propto \frac{M}{L} \propto M^{-3} \implies t_* = t_\odot \left(\frac{M}{M_\odot}\right)^{-3} \tag{8.20}$$

3.3 Convection

Convection becomes important if

$$t_{\mathrm{rad}} \gg t_{\mathrm{dyn}} = \frac{l}{c_s} \tag{8.21}$$

A more concrete criteria for convection

Let's look at a element of gas with density ρ and temperature T at radius r in a star, which moves to $r + dr$ and its parameters change to $\rho + \delta\rho$ and $T + \delta T$. The criterion for convection is

$$\rho_{\mathrm{element}} + \delta\rho_{\mathrm{element}} < \rho_{\mathrm{env}} + d\rho_{\mathrm{env}} \tag{8.22}$$

or

$$\delta\rho < d\rho \tag{8.23}$$

then the element will float upwards. For a more accurate derivation we'l assume this is an adiabatic process, which lets us say

$$P \propto \rho^\gamma \qquad\qquad \frac{\delta\rho}{\rho} = \frac{1}{\gamma}\frac{\delta P}{P} = \frac{1}{\gamma}\frac{dP}{P} \tag{8.24}$$

we'll also assume there is a pressure equilibrium with the star. Therefore

$$P = \frac{\rho}{m}\kappa T \tag{8.25}$$

which gives us

$$\frac{d\rho}{\rho} = \frac{dP}{P} - \frac{dT}{T} \tag{8.26}$$

We can now insert both 8.24 and 8.26 into 8.23 we find

$$\frac{1}{\gamma}\frac{1}{P}\frac{\mathrm{d}P}{\mathrm{d}r} < \frac{1}{P}\frac{\mathrm{d}P}{\mathrm{d}r} - \frac{1}{T}\frac{\mathrm{d}T}{\mathrm{d}r} \tag{8.27}$$

rearranging

$$\frac{\mathrm{d}T}{\mathrm{d}r} < \frac{\gamma - 1}{\gamma}\frac{T}{P}\frac{\mathrm{d}P}{\mathrm{d}r} \tag{8.28}$$

both the differentials are smaller than 0, so we'll multiply both sides by -1 and find

$$\left|\frac{\mathrm{d}T}{\mathrm{d}r}\right| > \frac{\gamma - 1}{\gamma}\frac{T}{P}\left|\frac{\mathrm{d}P}{\mathrm{d}r}\right| \tag{8.29}$$

The effect this has is that it improves the transfer of heat which in turn balances it out recursively, and after a while the greater symbol will become equal

$$\left|\frac{\mathrm{d}T}{\mathrm{d}r}\right| = \frac{\gamma - 1}{\gamma}\frac{T}{P}\left|\frac{\mathrm{d}P}{\mathrm{d}r}\right| \tag{8.30}$$

which is a stable state. If 8.29 is satisfied, we replace the radiation diffusion equation for the structure of stars with the convection equation 8.30.

Lecture 9.

Thu, November 27th, 2025

I missed this lecture. The presentation "Supernova, Cosmic-rays and Turbulence " was covered in this lecture.

Lecture 10.

Mon, December 1st, 2025

I missed this lecture.

Lecture 11.

Thu, December 4th, 2025

I missed this lecture. This lecture and the previous one covered pages 61 to 80 in Shmuel's handwritten notes. The notes I'm including here for this lecture are copied from those notes as closely to word for word as I could, and are not based on the lecture itself.

4 Remnants of Stars - Exotic Objects

Stars on the main sequence towards the end of their lives will become either Red Giants (If their mass is below 8 solar masses) or Supergiants (If their mass is above 8 solar masses). Red Giants then become Planetary Nebulas and eventually White Dwarfs. Supergiants collapse and explode in a Supenova and then end up as either a Neutron Star or a Black Hole.

4.1 White Dwarfs

In the 19-20 centuries it was determined that the nearby star Sirius (the brightest star in the night sky) is actually a pair of two stars, Sirius A and B, which is a white dwarf. From Kepler's law it was determined that Sirius B's mass is approximately 1 solar mass, but because it had low luminosity and high temperature, it was concluded that its radius is approximately 6000 km. This means its average density is

$$\overline{\rho} = \frac{M}{\frac{4\pi}{3}R^3} = 2\times 10^6\,\mathrm{g/cm^3} = 2\,\mathrm{ton/cm^3} \tag{11.1}$$

in other words, a teaspoon of a white dwarf is as heavy as an elephant. At the end of the star's life the core shrinks (because the fusion of hydrogen stops). Therefore both the density and temperature of the core grow, and as a result Helium and Carbon begin to fuse in the processes

$$3\,\mathrm{He}^4 \longrightarrow \mathrm{C}^{12} + \gamma \qquad\qquad \mathrm{He}^4 + \mathrm{C}^{12} \longrightarrow \mathrm{O}^{16} + \gamma \tag{11.2}$$

the first process being called "triple alpha" which releases 7.275 MeV. Eventually the Helium is used up which further compresses and heats the core, but eventually something stops the shrinking. Pauli's exclusion principle forbids electrons from coming too close together.

Which density does this start at?

When the distance between electrons become of the order of magnitude of $\frac{1}{2}\lambda_{DB}$ the wavefunctions star to 'feel' each other. In terms of density,

$$\rho_q \simeq \frac{m_p}{\left(\frac{1}{2}\lambda_{DB}\right)^3} = \frac{8m_p\left(3m_e k_B T\right)^{3/2}}{h^3} = 10^4 T_8^{3/2}\mathrm{g/cm^3} \tag{11.3}$$

where $T_8 \equiv \frac{T}{10^8\,\mathrm{K}}$. Once $\rho \gtrsim \rho_q$ the 'magic' happens. For the core of the sun, $T_8 = 0.015$ which gives $\rho_q = 640\,\mathrm{g/cm^3}$ but since the sun's core's density is $150\,\mathrm{g/cm^3}$ it is very far from the quantum limit. But when a star's core shrinks it is possible to reach it.

At The Quantum Limit

Particles' energies are distributed either by the Bose-Einstein distribution (for photons, gluons, etc) or the Fermi-Dirac distribution (for elcetrons, protons, neutrons, etc). The FD distribution at temperatures $k_B T \ll \epsilon_f$ it is approximately a step function where all energies $\epsilon \le \epsilon_f$ are populated and the rest are not. The fermi energy is

$$\epsilon_f = \frac{\hbar^2}{2m_e}\left(3\pi^2 n_e\right)^{2/3} \tag{11.4}$$

The lecture notes here develop the fermi energy, I skip it as it's included in the prerequesite course Statistical Mechanics. In conclusion, we'll find particles with high energy and momentum regardless of the temperature. They're being forced to be such by Pauli's exclusion principle.

The Pressure

Earlier in the course we developed the equation

$$P = \frac{1}{3}\int_0^\infty \mathrm{d}p p v \left(\frac{\mathrm{d}n}{\mathrm{d}p}\right) \tag{11.5}$$

if we assume the particles are non-relativistic we can substitute the velocity

$$P_e = \frac{1}{3m_e}\int_0^\infty \mathrm{d}p p^2 \left(\frac{\mathrm{d}n_e}{\mathrm{d}p}\right) \tag{11.6}$$

the number of electrons as a function of their momentum is

$$n_e = \frac{8\pi}{3}\left(\frac{p}{h}\right)^3 \tag{11.7}$$

so the density is

$$\frac{\mathrm{d}n_e}{\mathrm{d}p} = \begin{cases} 8\pi\frac{p^2}{h^3} & p \le p_f \\ 0 & p > p_f \end{cases} \tag{11.8}$$

inserting this back into 11.6

$$P_e = \frac{1}{3m_e}\int_0^{p_f} \mathrm{d}p p^2 \cdot 8\pi\frac{p^2}{h^3} \tag{11.9}$$

$$= \frac{8\pi}{3m_e h^3}\int_0^{p_f} \mathrm{d}p p^4 = \frac{8\pi}{15 m_e h^3} p_f^5 \tag{11.10}$$

inserting p_f

$$\boxed{P_e = \left(\frac{3}{8\pi}\right)^{2/3} \frac{h^2}{5m_e} n_e^{5/3}} \tag{11.11}$$

To find the expression in terms of density of mass we notice that

$$\rho = n_+ \cdot A m_p \tag{11.12}$$

and if we assume the material is fully ionized we can write $n_e = n_+ \cdot Z$ which gives us

$$\rho = n_+ A m_p = \frac{n_e}{Z} A m_p = m_p n_e \frac{A}{Z} \tag{11.13}$$

Therefore

$$\boxed{n_e = \frac{\rho}{m_p}\frac{Z}{A}} \tag{11.14}$$

inserting this into 11.9

$$\boxed{P_e = \left(\frac{3}{\pi}\right)^{2/3} \frac{h^2}{20 m_e m_p^{5/3}} \left(\frac{Z}{A}\right)^{5/3} \rho^{5/3}} \tag{11.15}$$

For a typical White Dwarf, $P_e \approx 3 \times 10^{22}\,\mathrm{dyne/cm^2}$ which is 5 orders of magnitude more than the sun's core.

An Estimate For a Typical Radius

Using the hydrostatic equation

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{GM_r}{r^2}\rho \tag{11.16}$$

and approximating

$$P \propto r^{\beta} \implies \frac{\mathrm{d}P}{\mathrm{d}r} \sim \frac{P}{r} \implies P \approx \frac{GM_\rho}{r} \tag{11.17}$$

$$\rho \approx \frac{M}{r^3} \implies \approx \frac{GM^2}{r^2} \tag{11.18}$$

Some development leads to

$$\boxed{r_{WD} = \frac{h^2}{20m_e m_p^{5/3} G}\left(\frac{Z}{A}\right)^{5/3} M^{-1/3}} \tag{11.19}$$

inserting constants

$$\boxed{r_{WD} \approx 10^9\,\mathrm{cm} \cdot \left(\frac{Z}{A}\right)\left(\frac{M}{M_\odot}\right)^{-1/3}} \tag{11.20}$$

What is P_e at the relativistic limit?

Inserting $v = c$ into 11.5 we find

$$\boxed{P_e = \left(\frac{3}{8\pi}\right)^{1/3} \frac{hc}{4m_p^{4/3}} \left(\frac{Z}{A}\right)^{4/3} \rho^{4/3}} \tag{11.21}$$

to find the mass-radius relation in the relativistic case we'll equate the hydrostatic pressure to the degeneracy pressure and find

$$\frac{GM^2}{r^4} \approx d\rho^{4/3} \tag{11.22}$$

if we insert the full expression for the density we'll find that the radius cancels out! This means we can't have an expression to connect the radius and mass! We find only a condition on the mass:

$$\boxed{M_{ch} = 0.21\left(\frac{hc}{Gm_p}\right)^{3/2}\left(\frac{Z}{A}\right)^2 m_p} \tag{11.23}$$

This is called the Chandrasekhar mass, and it is the maximum allowed mass for a white dwarf. If $M > M_{ch}$ the degeneracy pressure doesn't hold it together and it will collapse further. It's interesting to note that the Chandrasekhar mass has constants from different fields of physics.

Cooling of White Dwarfs

When White Dwarfs are created they are very hot ($T \sim 10^9\,\mathrm{K}$), and they cool down through black body radiation. To find the time it takes to cool down we can equate the luminosity of the star to the time derivative of the ideal gas law

$$-\sigma T^4 \cdot 4\pi r^2 = \frac{M}{m_p} k_B \frac{\mathrm{d}T}{\mathrm{d}t} \tag{11.24}$$

Therefore

$$t_{cool} = \int_0^{t_{cool}} \mathrm{d}t \tag{11.25}$$

$$= \frac{(M/m_p)\,k_B}{4\pi r^2 \sigma} \int_{T_0}^{T_f} \frac{\mathrm{d}T}{T^4} \tag{11.26}$$

$$= -\frac{(M/m_p)\,k_B}{4\pi r^2 \sigma} \left(-\frac{1}{3}\right)\left(\frac{1}{T_f^3} - \frac{1}{T_0^3}\right) \tag{11.27}$$

setting the initial inverse temperature cubed to be negligible we find

$$t_{cool} = \frac{(M/m_p)\,k_B}{12\pi r^2 \sigma} \frac{1}{T_f^3} \tag{11.28}$$

so the time it would take a white dwarf to cool to 1000 kelvin would be

$$t_{cool} = 15\,\mathrm{Gyr}\left(\frac{M}{M_\odot}\right)\left(\frac{4}{4000\,\mathrm{km}}\right)^{-2} \tag{11.29}$$

which means it'd take longer than the age of the universe. In practive the cooling time is even longer. For this reason white dwarfs appear white (or teal).

4.2 What Happens To More Massive Stars ($M > 8M_\odot$)?

After the fusion of hydrogen ends, the pressure goes down, their core shrinks, it gets hotter, and fusion of the next element begins. First the triple alpha reaction, until it runs out too and the process happens again. The next stage is fusion of Carbon into Oxygen and Oxygen into Neon

$$\mathrm{He}^4 + \mathrm{C}^{12} \longrightarrow \mathrm{O}^{16} + \gamma \qquad \mathrm{He}^4 + \mathrm{O}^{16} \longrightarrow \mathrm{Ne}^{20} + \gamma \tag{11.30}$$

$$2\,\mathrm{C}^{12} \longrightarrow \mathrm{Ne}^{20} + \mathrm{He}^4 + \gamma \qquad 2\,\mathrm{C}^{12} \longrightarrow \mathrm{Ne}^{23} + \mathrm{p} \qquad 2\,\mathrm{C}^{12} \longrightarrow \mathrm{Mg}^{23} + \mathrm{n} \tag{11.31}$$

and so on. Each stage is shorter than the last this continues until it reaches Iron. From this point

H	5 Myr
He	0.5 Myr
C	500 yr
Ne	1 yr
Si	1 day

fusion no longer takes place, which destabilizes the core. There is no source of pressure which causes the core to collapse and heat up. This causes two processes to take place, one of which is called photodisintegration, which is the process

$$\gamma^+\,\mathrm{Fe} \longrightarrow 13\,\mathrm{He} + 4\,\mathrm{n} \qquad \gamma^+\,\mathrm{He} \longrightarrow 2\,\mathrm{p} + \mathrm{n} \tag{11.32}$$

the first process consumes 124 MeV and the second 28 MeV. If we assume a white dwarf converted all its mass to iron the number of iron atoms is

$$N_{Fe} = \frac{M_{ch}}{m_{Fe}} \sim 3 \times 10^{55} \tag{11.33}$$

So the total energy to be lost is

$$E_{loss} \sim 2 \times 10^{52}\,\mathrm{erg} \tag{11.34}$$

which is a massive amount of energy, equivalent to the total energy radiated by the sun through its entire life.

Lecture 12.

Mon, December 8th, 2025

After both processes our star's material is rich in neutrons, which are fermions, and therefore Pauli's exclusion principle applies to them too. Everything is the same as for elections, the only difference being that they have a different λ_{DB}. A similar analysis to electrons can be done with two differences: using the neutron mass instead of the electron's, and setting $Z/A = 1$. This results in

$$R_{ns} = 2.3 \times 10^9 \frac{m_e}{m_n} \left(\frac{M}{M_\odot}\right)^{-1/3} = 11\,\mathrm{km} \left(\frac{M}{M_\odot}\right)^{-1/3} \tag{12.1}$$

which is about 500 times smaller than a white dwarf. The density in comparison to a whie dwarf will be

$$\frac{\rho_{ns}}{\rho_{wd}} \sim 500^3 \sim 10^8 \implies \rho_{ns} \sim 10^{14}\,\mathrm{g/cm^3} \tag{12.2}$$

If we look at the density of material in the nucleon of an atom it is

$$\rho_{nuc} = \frac{m_p}{\frac{4\pi}{3}(1\,\mathrm{fm})^3} \sim 4 \times 10^{14}\,\mathrm{g/cm^3} \tag{12.3}$$

So the density of a neutron star is in comparable to that of the nucleus of an atom. You could think of the entire thing as a single atom nucleus with atomic number 10^{57}. Recalling the expression for M_{ch}:

$$M_{ch} = 0.21 \left(\frac{Z}{A}\right)^2 \alpha_G^{-3/2} m_p \tag{12.4}$$

setting the relation we had for neutron stars

$$M_{ns} = 0.21 \alpha_G^{-3/2} m_p = 5.6 M_\odot \tag{12.5}$$

which is the upper limit for the mass of a neutron star. A neutron star more massive than that won't hold itself together and collapse further. Observations indeed haven't found any neutron stars heavier than (or even close to) that mass. More accurate calculations give a result significantly lower. The reason for this is that general relativistic effects become important at this scale. Relativistic effects become relevant when

$$\frac{E_{Gr}}{mc^2} \sim \frac{\frac{GM_{ns}m}{R_{ns}}}{mc^2} = \frac{GM_{ns}}{R_{ns}c^2} \sim 0.2 \tag{12.6}$$

Another important effect we haven't mentioned yet is the strong nuclear force. Since the neutron star effectivelty behaves like a giant nucleus, the strong force will start to have an effect. We don't quite understand what its effect is though. If we want to look at how much energy is expected to be released in the collapse of a core into a neutron star

$$\Delta E_{gr} = \left| \frac{\frac{3}{5}GM^2}{R_0} - \frac{\frac{3}{5}GM^2}{R_{ns}} \right| \approx \frac{\frac{3}{5}GM^2}{R_{ns}} \tag{12.7}$$

and if we insert typical quantities for neutron stars

$$M = 1.3 M_\odot \qquad\qquad R_{ns} = 11\,\mathrm{km} \tag{12.8}$$

we find

$$\Delta E_{gr} \sim 3 \times 10^{53}\,\mathrm{erg} \tag{12.9}$$

This energy has to go somewhere and this causes a supernova. This is the leading process which causes supernovas. This kind of supernova is called a core-collapse supernova. The luminosity we see from this process is and the total energy we observe is

$$E_\gamma = \int \mathrm{d}t L = 3 \times 10^{49}\,\mathrm{erg} \tag{12.10}$$

which is only 0.01% of the available energy. Where does the rest of the energy go? Maybe it is ejected from the supernova. The ejected mass is about $(1-10)\,M_\odot$ and observations show that its velocity is $v \sim 10^4\,\mathrm{km\,s^{-1}}$. This gives us an energy of

$$E_k \sim \frac{1}{2} M_{ej} v^2 = 3 \times 10^{51}\,\mathrm{erg} \tag{12.11}$$

which is 1% of the total energy, so there must be another explanation. Approximately 99% is ejected as neutrinos. The density in the center of the supernova is extremely high and the temperature is very high, which means there are very many photons whizzing around close to each other, and some of them collide with each other which results in

$$\gamma + \gamma \to e + e^+ \to \begin{cases} \nu_e + \overline{\nu}_e \\ \nu_\mu + \overline{\nu}_\mu \\ \nu_\tau + \overline{\nu}_\tau \end{cases} \tag{12.12}$$

The conditions are so extreme that sometimes even the neutrinos go through collision processes. The reason this process doesn't happen in the sun is that there just isn't enough energy there. The rest energy of the electron is $m_e c^2 = 0.5\,\mathrm{MeV}$ which photon collisions can't reach in the sun. Other types of explosions: Type 1a Supernovas. There's a white dward just chilling on its own but a red giant approaches it and the white dwarf starts sucking material out of the red giant. It then eventually reaches M_{ch} and explodes. In regular stars there is a negative feedback loop where fusion causes temperature to increase, which increases the pressure, which increases the radius of the star, which lowers the temperature, which lower the rate of fusion. In contrast in a white dwarf fusion causes an increase in temperature which doesn't affect the pressure, so it increases the rate of fusion again, and so on. This is called a type 1a supernova. Another kind of explosion is a gamma ray burst. The energy released from GRBs is $E \sim 10^{51}$ erg which is all released over about 0.1 seconds to a minute. GRBs are an active research area and it's theorized that there are several types.

Lecture 13.

Mon, December 15th, 2025

4.3 Pulsars

Pulsars are supernova remnants which appear to have pulsing fluxes of radaition. For examples, at the center of the Crab Nebula there is a pulsar which pulses with every 33 miliseconds. The current theory is that these pulses come from neutron stars with magnetic fields which rotate very rapidly. These magnetic fields create jets that go in opposite directions. We only observe these as pulsars if the jets point at us.

Why do they spin so fast?

If we look at a typical netron star of radius 10 km, and if it spins with period of 1 ms, then the speed of its surface is as fast as $0.2c$. Where does that speed come from? Conservation of angular momentum. Angular momentum for a sphere is

$$J = I\omega = \frac{2}{5}MR^2\omega \tag{13.1}$$

to satisfy conservation of angular momentum we have

$$J_0 = J_1 \tag{13.2}$$

$$\eta M R_1^2 \omega_1 = \eta M R_0^2 \omega_0 \tag{13.3}$$

$$\implies \frac{\omega_1}{\omega_0} = \left(\frac{R_1}{R_0}\right)^{-2} \tag{13.4}$$

for example, plugging in the sun's radius for the initial and a neutron star radius for the final radius, then we have

$$\left(\frac{R_1}{R_0}\right)^{-2} = 5 \times 10^9 \tag{13.5}$$

and since the sun's rotation's period is $\tau_0 = 25\,\mathrm{d} = 2 \times 10^6\,\mathrm{s}$, we have

$$\tau_1 = \tau_0 / 5 \times 10^9 \sim \mathrm{ms} \tag{13.6}$$

Where does the magnetic field come from?

The magnetic field of a pulsar is of approximately $10^{13}\,\mathrm{G}$. In comparison, the sun's magnetic field strength is $1\,\mathrm{G}$, and the earth's is $10^{-5}\,\mathrm{G}$. We can think of a pulsar as a rotating magnetic dipole. The radiation from this dipole is given by

$$L = \frac{1}{6c^3} B^2 R^6 \omega^4 \sin^2\theta \tag{13.7}$$

For the pulsar in the crab nebula we measure

$$L = 5 \times 10^{38}\,\mathrm{erg\,s^{-1}} \tag{13.8}$$

$$\omega = 190\,\mathrm{s^{-1}} \tag{13.9}$$

and if we assume a typical 10 km radius, we find $B \approx 10^{13}\,\mathrm{G}$.

How does the magnetic field get so strong?

During the collapse process, another process called flux freezing takes place. The magnetic flux is defined by, for stars

$$\Phi_B \propto BA = B \cdot \pi R^2 \tag{13.10}$$

doing a similar conservation of flux calculation we find

$$B_0 R_0^2 = B_1 R_1^2 \tag{13.11}$$

$$B_1 = B_0 \left(\frac{R_1}{R_0}\right)^{-2} \tag{13.12}$$

which fits our comparison earlier to the sun, if we plug in the numbers.

Observation of the Crab Nebula

Observation of the crab nebulas show that the frequency of pulses from the pulsar in the crab nebula today is

$$\omega_0 = 190\,\mathrm{s}^{-1} \tag{13.13}$$

and the rate of change in the frequency is

$$\dot{\omega} = 2.4 \times 10^{-9}\,\mathrm{s}^2 \tag{13.14}$$

so it's slowing down. To see why it's slowing down, we'll look at the energy it loses through radiation

$$\frac{\mathrm{d}E}{\mathrm{d}t} \propto \omega^4 \tag{13.15}$$

on the other hand

$$E = I\omega^2 \implies \frac{\mathrm{d}E}{\mathrm{d}t} = \omega\dot{\omega} \tag{13.16}$$

equating the two we find

$$\dot{\omega} = c\omega^3 \tag{13.17}$$

after solving we find

$$t = \frac{1}{2c}\left(\frac{1}{\omega_o^2} - \frac{1}{\omega_i^2}\right) \tag{13.18}$$

and with a little rearranging we have

$$t = \frac{\omega_0^3}{\dot{\omega}_0}\left(\frac{1}{\omega_0^2} - \frac{1}{\omega_i^2}\right) \tag{13.19}$$

if we assume significant deceleration $\omega_0 \ll \omega_i$

$$t = \frac{\omega_0}{\dot{\omega}_0} \tag{13.20}$$

inserting the numbers we observed we find

$$t = 1260\,\mathrm{yr} \tag{13.21}$$

4.4 Black Holes

If the mass of the star is greater than the maximum mass allowed for neutron stars it becomes a black hole. This gives us a lot of mass concentrated in a very small radius.

What is the escape velocity?

The kinetic energy of an object is

$$E_k = \frac{1}{2}mv^2 \tag{13.22}$$

while the gravitational potential energy is

$$E_G = \frac{GMm}{R} \tag{13.23}$$

equating the two we find

$$v_{esc} = \sqrt{\frac{2GM}{R}} \tag{13.24}$$

if we equate this to the speed of light we find

$$R_s = \frac{2GM}{c^2} \tag{13.25}$$

which means that if R, R_s even light cannot escape the star's surface. This is called a black hole. Some examples for Scwartzchild radii are this calculation is technically incorrect because we ignored

R_s	M
0.9 cm	$M_\oplus = 6 \times 10^{27}$ g
3 km	$M_\odot = 2 \times 10^{33}$ g
0.08 au	$M = 4 \times 10^6 M_\odot$
8 au	$4 \times 10^8 M_\odot$

the effects of relativity. The results are correct mostly by accident.

5 General Relativity

The Einstein equation is

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{13.26}$$

where $T_{\mu\nu}$ is the momentum-energy tensor and it describes the 'sources' and $G_{\mu\nu}$ describes the curvature of spacetime. $G_{\mu\nu}$ is comprised of the metric $g_{\mu\nu}$ and its derivatives. The simplest metric is the Minkowski metric which describes flat spacetime. We work with mostly minus, therefore the metric looks like

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{13.27}$$

and the four-position is defined by

$$\mathrm{d}x^\mu = \begin{cases} \mathrm{d}x^0 = c\,\mathrm{d}t \\ \mathrm{d}x^1 = \mathrm{d}x \\ \mathrm{d}x^2 = \mathrm{d}y \\ \mathrm{d}x^3 = \mathrm{d}z \end{cases} \tag{13.28}$$

As we know many quantities aren't invariant to lorentz boosts, e.g lengths and time. But there are some quantities which are invariant. An important lorentz invariant is the 'interval' $\mathrm{d}s$:

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu \tag{13.29}$$

this value can be either positive, negative, or zero. We can look at this through a figure. the volume inside the cone represents events which are causally linked. Massive objects could have moved between those two events, and for those events the interval is positive and a different frame of refence can be found in which both events happen at the same location. These events are called timelike. There are also events outside the cone. These events are not causally linked so no information can reach them. For them the interval is negative. A different frame of reference can be found in which both events happen at the same time. These events are called spacelike. The final kind of events are those directly on the cone. These events are called lightlike and for these events they are causally linked only for massless particles and the interval is equal 0.

5.1 The Minkowski Metric in Spherical Coordinates

We know what the metric looks like in cartesian coordinates, and we know the transformation to spherical coordinates, therefore

$$ds^2 = (c\,\mathrm{d}t)^2 - \mathrm{d}r^2 - (r\,\mathrm{d}\theta)^2 - (r\sin\theta\,\mathrm{d}\varphi)^2 \tag{13.30}$$

therefore the metric is

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -r^2 & 0 \\ 0 & 0 & 0 & -r^2\sin^2\theta \end{pmatrix} \tag{13.31}$$

Lecture 14.

Mon, December 22nd, 2025

If our space does have a mass M at the position $r = 0$ then the metric for the space is the Schwartzchild metric (in spherical):

$$g_{\mu\nu} = \begin{pmatrix} 1 - \frac{2GM}{rc^2} & & & \\ & -\left(1 - \frac{2GM}{rc^2}\right)^{-1} & & \\ & & -r^2 & \\ & & & -r^2 \sin^2 \theta \end{pmatrix} \tag{14.1}$$

where empty places are zeroes. We'll define the time t to be measured at infinity, therefore the interval is

$$ds^2 = dx^\mu dx^\nu g_{\mu\nu} \tag{14.2}$$

$$= \left(1 - \frac{2GM}{rc^2}\right)(cdt)^2 - \left[\left(1 - \frac{2GM}{rc^2}^{-1} dr^2 + (rd\theta)^2 + (r\sin\theta d\varphi)^2\right)\right] \tag{14.3}$$

If we define $\frac{2GM}{c^2} = R_s$ then

$$ds^2 = \left(1 - \frac{R_s}{r}\right)(cdt)^2 - \left[\left(1 - \frac{R_s}{r}\right)^{-1} dr^2 + \ldots\right] \tag{14.4}$$

and we can use this to define the proper time

$$\tau = \frac{ds}{c} \tag{14.5}$$

Let's look at the following scenario: we have a mass M at position $r = 0$, and our friend Yossi is located at position $\mathbf{r}$. We're located at infinity. we measure a time interval dt. What time interval τ did Yossi measure? We look at the interval between the two events. both events happened at the same position so $dr = d\theta = d\varphi = 0$. This means that

$$ds^2 = \left(1 - \frac{R_s}{r}\right)(cdt)^2 \tag{14.6}$$

therefore

$$\tau = \frac{ds}{c} = \sqrt{1 - \frac{R_s}{r}} dt \tag{14.7}$$

if we define $\Delta r = r - R_s$ then

$$d\tau = dt\sqrt{\frac{\Delta r}{r}} \tag{14.8}$$

and if we look at some numerical examples

$\frac{\Delta r}{r}$	$\frac{\Delta \tau}{\tau}$
0.5	1/1.4
0.1	~ 1/3
0.01	1/10
10^{-4}	1/100

Let's give Yossi a flashlight and throw him into a black hole. We could have thrown the flashlight on its own, but that would be less fun. What we'll see is that Yossi's radius asymptotically approaches R_s but never crosses it. If we look at what the flashlight does, we'll notice the wavelength of the light increases and the magnitude of the light decreases. If we define the period of the wavelength of the light to be T_0 and the frequency to be ν_0 then the period at infinity is

$$T_\infty = \frac{1}{\sqrt{1 - \frac{R_s}{r}}} T_0 \tag{14.9}$$

and

$$\nu_\infty = \nu_0 \sqrt{1 - \frac{R_s}{r}} \tag{14.10}$$

and the wavelength would be

$$\lambda_\infty = \frac{\lambda_0}{\sqrt{1 - \frac{R_s}{r}}} \tag{14.11}$$

if we look at a source of light which sends pulses of photons with frequency ν_0 at the rate $\kappa(s^{-1})$ then the power output of it is

$$P_0 = h\nu_0\kappa \tag{14.12}$$

and at infinity

$$P_\infty = P_0 \left(1 - \frac{R_s}{r}\right) \tag{14.13}$$

Let's now look at the behavior of light near black holes. To do that we'll look at the case of the interval being zero, $ds = 0$:

$$0 = \left(1 - \frac{R_s}{r}\right)(cdt)^2 - \left(1 - \frac{R_s}{r}\right)^{-1} dr^2 + dots \tag{14.14}$$

for a radial path $d\theta = d\varphi = 0$and we'll find

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \pm\left(1 - \frac{R_s}{r}\right)c \tag{14.15}$$

but does this not violate the principle that light moves at the speed of light? No. This is the speed we observe as an external observer at infinity. the velocity as experienced by the light itself is

$$v = \frac{\mathrm{d}l}{\mathrm{d}\tau} = \frac{\mathrm{d}r}{\mathrm{d}t} \frac{\frac{1}{\sqrt{1 - \frac{R_s}{r}}}}{\sqrt{1 - \frac{R_s}{r}}} = \frac{\frac{\mathrm{d}r}{\mathrm{d}t}}{1 - \frac{R_s}{r}} = c \tag{14.16}$$

5.2 Thermodynamics of Black Holes

[1] The 2nd law of thermodynamics states that entropy increases, i.e $ds \geq 0$. The entropy of a black hole is proportional to it surface area $S_{BH} \propto A_{BH}$. If mass is absorbed by a black hole, its mass increases. This in turn increases the radius which increases the surface area, and thus the entropy increases. A more accurate calculation done by Hawking and Bekerstein yields

$$S_{BH} = k_B \frac{A}{4l_p^2} \tag{14.17}$$

where

$$l_p = \sqrt{\frac{G\hbar}{c^3}} \tag{14.18}$$

for a Schwartzchild black hole, $A = 4\pi R_s^2$. The black hole having an entropy which changes when mass is absorbed implies that it might have a definable temperature, which itself implies that we may

[1] This is a very complex subject so we won't go deep into it but it's very interesting so we'll briefly cover it. Some interesting subtopics we won't cover are the information paradox, the Unruh effect, and Hologram Universe.

be able to measure a radaition being emitted from the black hole as a black body. We can define temperature using statistical mechanics to be

$$T = \frac{\mathrm{d}U}{\mathrm{d}S} = \frac{\mathrm{d}U}{\mathrm{d}M}\left(\frac{\mathrm{d}S}{\mathrm{d}M}\right)^{-1} \tag{14.19}$$

as we are looking at a Schwartzchild black hole we can define $U = Mc^2$ and find

$$S = k_B\frac{A}{l_p^2} = k_B\frac{4\pi R_s^2}{4l_p^2} = k_B\frac{\pi}{l_p^2}\left(\frac{2GM}{c^2}\right)^2 \tag{14.20}$$

therefore

$$\frac{\mathrm{d}S}{\mathrm{d}M} = k_B\frac{8\pi GM}{\hbar c} \tag{14.21}$$

and

$$\frac{\mathrm{d}U}{\mathrm{d}M} = c^2 \tag{14.22}$$

inserting back into 14.19

$$k_BT = \frac{\hbar c^3}{8\pi GM} \tag{14.23}$$

if we plug in numbers then the temperature is

$$T = 60\,\mathrm{nK}\left(\frac{M}{M_\odot}\right)^{-1} \tag{14.24}$$

now since the black hole has a temperature, it'll radiate as a black body. How long would it take for a black hole then to radiate away all its mass?

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{\mathrm{d}Mc^2}{\mathrm{d}t} = c^2\dot{M} \tag{14.25}$$

but on the other hand

$$\frac{\mathrm{d}E}{\mathrm{d}t} = 4\pi R_s^2\sigma T^4 \propto M^2\cdot\left(\frac{1}{M}\right)^4 \propto \frac{1}{M^2} \tag{14.26}$$

which gives us a differential equation whose solution is

$$t_{evap} = \frac{5120\pi G^2M^3}{\hbar c^4} \tag{14.27}$$

inserting the numbers find

$$t_{evap} = 2\times10^{67}\,\mathrm{yr}\left(\frac{M}{M_\odot}\right)^3 \tag{14.28}$$

so black holes practically never evaporate. For a black hole to evaporate within the age of the universe, they'd have to have mass

$$M \leq 10^8\,\mathrm{g} \tag{14.29}$$

Lecture 15.

Thu, December 25th, 2025

I missed this lecture. This lecture. The notes I'm including here for this lecture are copied from Shmuel's handwritten notes as closely to word for word as I could, and are not based on the lecture itself.

6 Intestellar Medium

The ISM includes

- Gas
- Dust
- Cosmic Rays
- EM Radiation
- Magnetic Fields
- Dark Matter

There isn't a property that is uniform across all the ISM, and the ISM is inhomogenous. Some areas are molecular, some are atomic, and some are ionized. Typical values for the temperature and density of the various states of matter are on average, there is about 1 particle of the above per

Chemical State	Temperature $[K]$	Density $[\text{cm}^3]$
Molecular	$10 - 100$	$100 - 10^6$
Atomic	$100 - 10^4$	$10^{-1} - 100$
Ionized	$10^4 - 10^7$	$10^{-4} - 10^{-1}$

cubic centimeter. For comparison, in Earth's atmosphere there are approximately $10^9\,\text{cm}^3$ particles. The densest areas collapse and create stars. Those regions are called 'Molecular Clouds' and their parameters are

$$M = 100 - 10^5 M_\odot \qquad n = 10^2 - 10^4\,\text{cm}^3 \qquad T \sim 10 - 100K \tag{15.1}$$

Even the most dense regions are still very far from the density of stars. The process of collapse is not completely understood and is an active area of research.

6.1 Hydrodynamics

To describe the behavior of these regions we'll describe particles in a statistical manner and we'll want to characterize them using their density $\rho(\mathbf{r}, t)$, their velocity $\mathbf{v}(\mathbf{r}, t)$, the pressure $P(\mathbf{r}, t)$, and the temperature $T(\mathbf{r}, t)$. The minimum free pass distance for the randomly moving particles is

$$\lambda_{mfp} = \frac{1}{\sigma n} \sim 10^{13}\,\text{cm} \tag{15.2}$$

The variables used to describe the system are described by a set of interconnected partial differential equations called the Navier-Stokes equations, or the Magneto-Hydrodynamics equations. The equations result from conservation laws, such as conservation of mass, conservation of momentum, and conservation of energy. There are several ways to develop the equations, we'll follow a (shortened) development which follows mass elements in a fluid from a lagrangian frame of reference (which moves with the element). To turn an ODE into a PDE we can use the following trick

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{\partial A}{\partial t} + \frac{\partial A}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial A}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial A}{\partial z}\frac{\partial z}{\partial t} = \left[\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla\right]A \tag{15.3}$$

where the expression $\frac{\mathrm{d}}{\mathrm{d}t} = \left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla\right)$ is called the convective/mass/material derivative.

Conservation of Mass: Continuity Equation

The first equation we'll look at is the equation for conservation of mass,

$$\frac{\mathrm{d}M}{\mathrm{d}t} = 0 \tag{15.4}$$

rewriting using the identity 15.3

$$\frac{\partial}{\partial t}\left(\rho dV\right) = -\rho\mathbf{v}\cdot d\mathbf{A} \tag{15.5}$$

recalling the divergence theorem

$$\oiint_S \rho\mathbf{v}\cdot\mathrm{d}\mathbf{A} = \iiint_V \nabla\left(\rho\mathbf{v}\right)\mathrm{d}V \tag{15.6}$$

we can thus rewrite 15.5 as

$$\frac{\partial\rho}{\partial t} = -\nabla\left(\rho\mathbf{v}\right) \tag{15.7}$$

which gives us

$$\frac{\partial\rho}{\partial t} + \nabla\left(\rho\mathbf{v}\right) = 0 \tag{15.8}$$

which is the continuity equation. For an incompressible gas (such as water) $\rho = const$ therefore $\rho\nabla\mathbf{v} = 0$ and thus the continuity equation becomes

$$\nabla\mathbf{v} = 0 \tag{15.9}$$

the ISM is sometimes very compressible, meaning shockwaves can happen.

Lecture 16.

Mon, December 29th, 2025

6.2 Conservation of Momentum

we reached the equation

$$\rho \delta V \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \sum \delta \mathbf{F_i} \tag{16.1}$$

where F_i is the parts that depend on the pressure, EM field, gravitation, and viscosity.

The pressure element δF_p is

$$\mathbf{F_p} = \iint -p \, \mathrm{d}\mathbf{s} = -\iint_V \nabla p \, \mathrm{d}V \tag{16.2}$$

$$\implies \delta \mathbf{F_p} = -\nabla p \delta V \tag{16.3}$$

the electromagnetic element $\delta \mathbf{F_B}$

Assuming a neutral plasma, we can find from the lorentz force on a particle

$$\mathbf{F} = \frac{q\mathbf{v}}{c} \times \mathbf{B} \tag{16.4}$$

and in the continuum limit it is

$$q\mathbf{v} \to \mathbf{J}\delta V \tag{16.5}$$

so

$$\mathbf{F_B} = \frac{\mathbf{J}}{c} \times \mathbf{B} \, \mathrm{d}V \tag{16.6}$$

$$\delta \mathbf{F_B} = \frac{\mathbf{J}}{c} \times \mathbf{B} \delta V \tag{16.7}$$

the maxwell equation relating $\mathbf{J}$ to $\mathbf{B}$ is

$$\nabla \times \mathbf{B} = \frac{4\pi}{c}\mathbf{J} \implies \mathbf{J} = \frac{c}{4\pi}(\nabla \times \mathbf{B}) \tag{16.8}$$

inserting this back into 16.7 we find

$$\delta \mathbf{F_B} = \frac{1}{4\pi}(\nabla \times \mathbf{B}) \times \mathbf{B} \tag{16.9}$$

using a the 'BAC-CAB' tensor rule

$$\delta \mathbf{F_B} = \frac{1}{4\pi}\left[-\frac{1}{2}\nabla\left(B^2\right) + (\mathbf{B}\cdot\nabla)\mathbf{B}\right]\delta V \tag{16.10}$$

The graitationa element δF_g

If we assume a potential $\Phi(\mathbf{r}, t)$ then

$$\delta \mathbf{F_g} = -\delta m \nabla \Phi \tag{16.11}$$

$$\delta \mathbf{F_g} = -\rho \nabla \Phi \delta V \tag{16.12}$$

The viscosity element δF_ν

A way to write this element is

$$F_{\nu i} = \int \sigma_{ij}\, \mathrm{d}A_j \tag{16.13}$$

where σ_{ij} is the viscous stress tensor. Specifically for liquids it is often written as

$$\sigma_{ij} = \mu\left[\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\nabla\cdot\mathbf{v}\right] \tag{16.14}$$

inserting back into 16.13 and using Gauss' law

$$\delta\mathbf{F}_\nu = \mu\left(\nabla^2\mathbf{v} + \frac{1}{3}\nabla\left(\nabla\cdot\mathbf{v}\right)\right)\delta V \tag{16.15}$$

Returning to 16.1

inserting all the elements back into 16.1 we find

$$\rho\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla\left(p + \frac{B^2}{8\pi}\right) + \frac{1}{4\pi}\left(\mathbf{B}\cdot\nabla\right)\mathbf{B} - \rho\nabla\Phi_g + \mu\left(\nabla^2\mathbf{v} + \frac{1}{3}\nabla\left(\nabla\cdot\mathbf{v}\right)\right) \tag{16.16}$$

Which is the Navier-Stokes equation for MHD. The left hand side can be expanded to partial derivatives.

6.3 Important Dimensionless Numbers

Mach Number

The isothermic speed of sound is given by

$$c_s^2 = \frac{P}{\rho} \tag{16.17}$$

and a number called the 'mach number' as

$$M_s = \frac{v}{c_s} \tag{16.18}$$

If we look at the ratio between the dynamic element of naviert stokes and the pressure element

$$\frac{|\rho\left(\mathbf{v}\cdot\nabla\right)\mathbf{v}|}{|\nabla p|} \sim \frac{\rho v^2}{\rho c_s^2} = \left(\frac{v}{c_s}\right)^2 = M_s^2 \tag{16.19}$$

If $M_s \ll 1$ then $v \ll c_s$ and disturbances have time to move through the liquid and it equalizes and reorganizes neatly. If $M_s \gg 1$ then $v \gg c_s$ and shockwaves are created in the gas.

Alfvenic Mach Number

If we look at the ratio

$$\left|\frac{\rho\left(\mathbf{v}\cdot\nabla\right)\mathbf{v}}{\nabla\left(\frac{B^2}{8\pi}\right)}\right| \sim \frac{\rho v^2}{\frac{B^2}{8\pi}} = \left(\frac{v}{\frac{B}{\sqrt{8\pi\rho}}}\right)^2 \tag{16.20}$$

The element

$$v_A = \frac{B}{\sqrt{8\pi\rho}} \tag{16.21}$$

is called the Alfvenic speed, and the ratio is thus

$$M_A^2 = \left(\frac{v}{\frac{B}{\sqrt{8\pi\rho}}}\right)^2 = \left(\frac{v}{v_A}\right)^2 \tag{16.22}$$

and is called the Alfvenic mach number. If $M_A \ll 1$ then the magnetic field is strong and it dampens the disturbances. If $M_A \gg 1$ then the magnetic field is weak.

Reynolds Number

The Reynolds number is defined as

$$R_e = \frac{vL}{\nu} \tag{16.23}$$

where

$$\nu = \frac{\mu}{\rho} \tag{16.24}$$

If we look at the ratio

$$\left|\frac{\rho\left(\mathbf{v}\cdot\nabla\right)\mathbf{v}}{\mu\nabla^2\mathbf{v}} \sim \frac{\rho v^2}{\mu\frac{1}{L}v} = \frac{\rho v^2 L}{\rho\nu v} = \frac{vL}{\nu}\right| = R_e \tag{16.25}$$

If R_e is small the viscosity is important. If R_e is small then viscosity is negligible.

Typical Values in ISM

$$M_S \sim 0.1 - 50 \qquad M_A \sim 0.3 - 50 \qquad R_e \sim 100 - 10^8 \tag{16.26}$$

Example - Molecular Cloud

We have a Giant Molecular Cloud whose typical parameters are

$$n = 10^3\,\mathrm{cm}^3 \qquad T = 20\,\mathrm{K} \qquad v_{rms} = 5\,\mathrm{km\,s^{-1}} \qquad L = 10\,\mathrm{pc} \qquad B = 30\,\mu\mathrm{G} \tag{16.27}$$

and the important numbers are

$$M_S = 10 \qquad M_A \simeq 5 \tag{16.28}$$

and in homework we'll prove we can write

$$R_e = \frac{L}{\lambda_{mfp}}\frac{v}{c_s} \tag{16.29}$$

which for a GMC we have

$$\lambda_{mfp} = \frac{1}{\sigma n} \qquad \sigma = 10^{-15}\,\mathrm{cm} \tag{16.30}$$

therefore

$$\lambda_{mfp} = 10^{12}\,\mathrm{cm} \tag{16.31}$$

and that gives us

$$R_e = 3\times10^8 \tag{16.32}$$

In ISM, sometimes the viscosity element is negligible so the NS equation is simplified to

$$\rho\frac{\partial\mathbf{v}}{\partial t} + \rho\left(\mathbf{v}\cdot\nabla\right)\mathbf{v} = -\nabla\left(p + \frac{B^2}{8\pi}\right) + \frac{1}{4\pi}\left(\mathbf{B}\cdot\nabla\right)\mathbf{B} - \rho\nabla\Phi_g \tag{16.33}$$

We can also use the state equation $P = nk_BT$. But what determined T? From conservation of energy

$$\mathrm{d}U = -p\,\mathrm{d}V + \mathrm{d}Q \tag{16.34}$$

where

$$U = \frac{f}{2}Nk_BT \tag{16.35}$$

If we look at $\mathrm{d}Q$

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = V\left[\Gamma - \Lambda + \nabla\left(\kappa\nabla T\right)\right] \tag{16.36}$$

where Γ is the rate of heating per unit volume, Λ is the rate of cooling per unit volume, and the last element is for the diffusion of heat. Inserting 16.35 and 16.36 into 16.34 we find

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\Gamma - \Lambda + \nabla\left(\kappa\nabla T\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(nk_BT\right)}{\frac{1}{2}\left(f+2\right)nk_B} \tag{16.37}$$

6.4 Simple Problems

Free Collapsing Cloud

We start with a cloud of radius R with mass which has nothing opposing its collapse. In this case we can write

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = -\nabla\Phi_g \tag{16.38}$$

which can be directly solved but there's an easier way to do this, using energy

$$E_0 = -\frac{GM}{R_0}\delta M \tag{16.39}$$

and the energy at a later point is

$$E = \frac{1}{2}\delta M v^2 - \frac{GM}{r}\delta M \tag{16.40}$$

setting both to be equal we find an ODE

$$\frac{1}{2}v^2 = GM\left(\frac{1}{r} - \frac{1}{R_0}\right) \tag{16.41}$$

which can be rewritten as

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \sqrt{2GM\left(\frac{1}{r} - \frac{1}{R_0}\right)} \tag{16.42}$$

whose solution is

$$\int_0^{t_{ff}} \mathrm{d}t' = \sqrt{\frac{R_0^3}{2GM}}\int_0^1 \mathrm{d}x\left(\frac{1}{x} - 1\right)^{-1/2} \tag{16.43}$$

where $x = r/R_0$ which gives us

$$t_{ff} = 1.1 \cdot \sqrt{\frac{R_0^3}{GM}} \propto \frac{1}{\rho_0^{1/2}} = \sqrt{\frac{3\pi}{32G\rho_0}} = n_3^{-1/2}\mathrm{Myr} \tag{16.44}$$

where

$$n_3 = \frac{n}{10^3\,\mathrm{cm}^3} \tag{16.45}$$

Lecture 17.

Thu, January 1st, 2026

I missed this lecture.

Lecture 18.

Mon, January 5th, 2026

6.5 Heating-Cooling

Most of the gas, in terms of mass, in the ISM is neutral atomic gas. The gas can be split into two categories

CNM Cold Neutral Medium, which has a temperature of about 100 K and density of approximately 30 cm^3

WNM Warm Neutral Medium, which has a temperature of about 10^4 K and density of approximately 0.3 cm^3

Why those specific temperatures and densities? The main cooling processes relevant are line cooling of C^+ and $H - Ly\alpha$.

6.6 Cooling by Lines

Looking at two energy levels 1 and 0 with an energy gap of ΔE, there are three processes which can take place: excitation by collision, decay by collision, and radiative decay.

Excitation by Collision

the rate at which the population of 1 grows is

$$\frac{\mathrm{d}n_1}{\mathrm{d}t} = \langle \sigma_{01} v \rangle n_0 n_{col} = K_{01} n_0 n_{col} \tag{18.1}$$

where n_0 is the density of the ground level, n_1 is the density of the atoms in the 1st excited level, and n_{col} is the density of the colliding particles, and the element in angled brackets is the rate coefficient.

Decay by Collision

$$\frac{\mathrm{d}n_1}{\mathrm{d}t} = -K_{10} n_1 n_{col} \tag{18.2}$$

Radiative Decay

$$\frac{\mathrm{d}n_0}{\mathrm{d}t} = A_{10} n_1 \tag{18.3}$$

where A is the einstein coefficient for spontaneous decay. The total rate of change is therefore

$$\frac{\mathrm{d}n_0}{\mathrm{d}t} = n_1 n_{col} K_{10} + n_1 A_{10} - n_0 n_{col} K_{01} \tag{18.4}$$

After a certain amount of time the equation will reach a steady state determined by the coefficients. In this state the ratio between the first excited state and the ground state is

$$\frac{n_1}{n_0} = \frac{n_{col} K_{01}}{n_{col} K_{10} + A_{10}} \tag{18.5}$$

At high densities the radiative element is negligible and we have only excitation and decay via collisions, and we have

$$\frac{n_1}{n_0} = \frac{K_{01}}{K_{10}} \tag{18.6}$$

and since we have two opposing processes we should be able to have a state of thermal equilibrium at which

$$\frac{n_1}{n_0} = \frac{g_1}{g_0} e^{-\frac{\Delta E}{k_B T}} \tag{18.7}$$

where g is the degeneracy of the level. We can rewrite equation 18.5 as

$$\frac{n_1}{n_0} = \frac{K_{01}}{K_{10}} \left(\frac{1}{1 + \frac{A_{10}}{n_{col} K_{10}}} \right) \tag{18.8}$$

$$= \frac{g_1}{g_0} e^{-\frac{\Delta E}{k_B T}} \left(\frac{1}{1 + \frac{A_{10}}{n_{col} K_{10}}} \right) \tag{18.9}$$

and if we define

$$n_{crit} = \frac{A_{10}}{K_{10}} \tag{18.10}$$

then

$$\frac{n_1}{n_0} = \frac{g_1}{g_0} e^{-\frac{\Delta E}{k_B T}} \left(\frac{1}{1 + \frac{n_{crit}}{n_{01}}} \right) \tag{18.11}$$

Which tells us that cooling is effective when $n_{col} \ll n_{crit}$. In that case the cooling rate is

$$\frac{n_1}{n_0} \simeq \frac{g_1}{g_0} e^{-\frac{\Delta E}{k_B T}} \frac{n_{col}}{n_{crit}} \tag{18.12}$$

$$= \frac{g_1}{g_0} e^{-\frac{\Delta E}{k_B T}} n_{col} \frac{K_{10}}{A_{10}} \tag{18.13}$$

On the other hand the radiative coooling rate is

$$L_{cool} = \frac{\mathrm{d}E}{\mathrm{d}t\,\mathrm{d}V} = n_1 A_{10} \tag{18.14}$$

$$= \frac{n_1}{n_0} A_{10} \Delta E n_0 \tag{18.15}$$

$$= \frac{g_1}{g_0} e^{-\frac{\Delta E}{k_B T}} n_{col} K_{10} n_0 \Delta E \tag{18.16}$$

So the rate of cooling can be written as

$$L_{cool} = L(T) + n^2 \tag{18.17}$$

In The Lyman Alpha Line

The energy gap is

$$\Delta E = 13.6\,\mathrm{eV} \cdot \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = 10.2\,\mathrm{eV} \tag{18.18}$$

therefore

$$\frac{\Delta E}{k_B} \approx 10^5\,\mathrm{K} \qquad A_{10} = 1\,\mathrm{s}^{-1} \qquad K_{10} = 5.31 \times 10^{-4}\,\mathrm{cm}^3/\mathrm{s} \qquad n_{crit} = 1800\,\mathrm{cm}^3 \tag{18.19}$$

In The C2 Line

Looking at a carbon atom which has an outer electron with spin up. In the transition from spin up to spin down there is an energy difference of

$$\Delta E = 7.9 \times 10^{-3}\,\mathrm{eV} \tag{18.20}$$

therefore

$$\frac{\Delta E}{k_B} = 91\,\mathrm{K} \qquad K_{10} = 10^{-9}\,\mathrm{cm}^3/\mathrm{s} \tag{18.21}$$

6.7 Heating

The dominant heating process is called Photoelectric heating by far-UV radiation. A uv photon with energy 6 to 13.6 eV is released from an O/B star and hits a dust cloud. This ionizes an atom, releasing an electron. This electron undergoes processes and heats the cloud. The expression for this is

$$G = \frac{\mathrm{d}E}{\mathrm{d}t\,\mathrm{d}V} \tag{18.22}$$

$$= An_{dust}n_{rad} \tag{18.23}$$

if we assume

$$b_{dust} \propto n \tag{18.24}$$

then

$$G = B \cdot n_{rad}n \tag{18.25}$$

6.8 Heating-Cooling Equilibrium

the expression is

$$L(T)n^2 = Cn \tag{18.26}$$

$$n = \frac{C}{L(T)} \tag{18.27}$$

Lecture 19.

Thu, January 8th, 2026

Today we did not do any new material, we were shown a bunch of cool ISM simulations.

Lecture 20.

Mon, January 12th, 2026

This lecture was given by Prof. Vincent Desjacques

7 Cosmology

We assume that the universe is infinite in space, euclidian, and static. This is a problematic assumption but we make it. We'll look at a very basic observation which raises doubts as to at least one of these assumptions. We ask the question: why is the sky black at night(This is called the Olbers paradox)? To look further into this question, we'll assume there is a certain uniform average density of stars in the universe n_*, and that each star has an average radius of R_*. We'll look at a spherical shell of radius r and width $\mathrm{d}r$. We ask the question of what is the total solid angle of the stars in this shell:

$$\mathrm{d}\Omega = 4\pi r^2 \,\mathrm{d}r \cdot n_* \cdot \frac{\pi R_*^2}{r^2} \tag{20.1}$$

$$= 4\pi^2 n_* R_*^2 \,\mathrm{d}r \tag{20.2}$$

Now to find the total solid angle of all the stars in the universe, we take an integral over all radii from us of the shell

$$\Omega = \int_0^\infty \mathrm{d}r \frac{\mathrm{d}\Omega}{\mathrm{d}r} \tag{20.3}$$

$$= 4\pi^2 n_* R_* \int_0^\infty \mathrm{d}r \tag{20.4}$$

and this integral diverges. Technically this calculation is incorrect because we didn't take into consideration the possibility of stars overlapping, and if we did, then the total angle would be $\Omega = 4\pi$ i.e the entire sky would be full of stars. If we define the angular luminosity as $S = \frac{f}{\Omega}$ where f is the radiation flux, then we can see it does not depend on r. This means the average angular brightness of the sky should be similar to that of the sun. The night should be as bright as the sun is. But it's not. This means at least one of the assumptions we made is incorrect. The incorrect assumption is that the universe is *not* static.

7.1 Measuremnts of Distances in The Universe

To measure distances to relatively nearby stars, in our galaxy, up to about 1 kpc, we can use the parallax method. Earth goes around the sun, so if we measure the direction to the star half a year apart, we'll have a different angle and using the distance of the earth to the sun and the angle we can find the distance. This is accurate up to several hundred parsecs.

If we leave the galaxy and look at distances of about ~ 10 Mpc, which is out local group and nearby galaxy clusters, then the method to measure distances for astronomical objects of these distances is using Cepheid stars, often called 'Standard Candles'. These stars have periodically varying luminosity over time, and there is a clear correlation between the star's luminosity and its period length. There are Cepheids near earth, so we can use them to calibrate then once we find distant cepheids we can measure its luminosity over time, find the period, then find its actual luminosity, and by comparing to the measured luminosity we can know the distance.

We zoom out further, and ask how we can measure distances of about ~ 100 Mpc. At these distances we can't see individual stars, only entire galaxies. There are two methods we can use. The first is called the 'Tully-Fisher' method, where we have a tie between the luminosity of a galaxy and the maximum velocity of the gas moving around it:

$$L \propto v_{\max}^{\alpha} \tag{20.5}$$

we can measure the velocity of the gas using the difference in redshift between the side that moves towards us and the side that moves away from us. α was measured and found empirically to be $\alpha \simeq 4$. We can then again measure the observed luminosity and find the distance. Another kind of standard candle are type Ia supernovae. They have a very specific maximum brightness. We can measure the observed maximum brightness and compared to the known actual brightness and use that. There are many more methods to measure distances but we won't cover them.

7.2 Hubble's Law

Hubbles' law states that aside from a certain small number of nearby galaxies, galaxies in the universe are moving away from us. To reach this law, hubble measured the redshift of galaxies to find their velocities. The definition of redshift is

$$z = \frac{\lambda_{obs} - \lambda_{em}}{\lambda_{em}} \tag{20.6}$$

where λ_{obs} is the observed wavelength and λ_{em} is the emitted wavelength. We can't measure the emitted wavelength directly obviously, but by assuming that physics is the same across the universe, then the spectral lines of hydrogen as we measure here must be the same as there. If this was wrong, I'd just quit my job and go do something else. If we can find the spectral lines from the observations and match them to spectral lines we know from experiments, we can find the redshift. Hubble then assumed the redshift was the result of the doppler effect:

$$\frac{\lambda_{obs}}{\lambda_{em}} = 1 + z = \sqrt{\frac{1+\beta}{1-\beta}} \simeq +\beta \tag{20.7}$$

where $\beta = \frac{v}{c}$ and the last transition being in the limit $v/c \ll 1$. This assumption was wrong and we'll see why. He then plotted the velocity over the distance from us From a linear fit to the measurements we have

$$v = H_0 d \qquad\qquad H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}} \tag{20.8}$$

which is called Hubble's Law. Unless we abandon the copernican principle, our conclusion is that the universe is not static.

7.3 Describing A Non-Static Universe

We'll abandon the assumption the universe is static, but we'll make another assumption. We'll assume the universe is homogenous and isotropic, which is called the cosmological principle. We'll define a set of coordinates

$$\mathbf{r}(t) \equiv a(t)\mathbf{x} \tag{20.9}$$

where $\mathbf{r}$ is the physical distance, a is the scale factor, and $\mathbf{x}$ is the comoving distance. The comoving distance is a set of coordinates indicating the relative location of the object in the universe subtracting the expansion, which is facotred in with the scale factor. An observer with a constant $\mathbf{x}$ is called a 'comoving observer'. The velocity of an object is

$$\mathbf{v}(\mathbf{r}, t) = \frac{\mathrm{d}\mathbf{r}(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(a(t)\mathbf{x}\right) \tag{20.10}$$

$$= \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t}\left(a\mathbf{x}\right) \tag{20.11}$$

$$= \frac{\dot{a}}{a}\mathbf{r}(t) \tag{20.12}$$

and the visible velocity, which is the velocity it looks like the object is moving at from the expansion of the universe, rather than it moving through space is

$$\Delta \mathbf{v} = \mathbf{v}(\mathbf{r} + \Delta\mathbf{r}, t) - \mathbf{v}(\mathbf{r}, t) \tag{20.13}$$
$$= \frac{\dot{a}}{a}(\mathbf{r} + \Delta\mathbf{r}) - \frac{\dot{a}}{a}\mathbf{r} \tag{20.14}$$
$$= \frac{\dot{a}}{a}\Delta\mathbf{r} \tag{20.15}$$

we can define $H(t) = \frac{\dot{a}(t)}{a(t)}$ which is called the Hubble Rate at time t, and Hubble's constant becomes $H_0 \equiv H(t_0)$. $H(t)$ measures the rate of expansion of the universe over time. *This is not the doppler effect. This is true for any comoving observer.*

7.4 Space-Time

In special relativity the metric is the Minkowski metric

$$\mathrm{d}s^2 = c^2 c^2\, \mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2 \qquad x^\mu = (t, x, y, z) \tag{20.16}$$

which is the metric of a static unchanging homogenous isotropic universe which is no good. The metric we'll use instead is

$$\mathrm{d}s^2 = c^2\, \mathrm{d}t^2 - a^2(t)\sigma_{ij}\, \mathrm{d}x^i\, \mathrm{d}x^j \tag{20.17}$$

This metric can describes three types of spaces, and we can define a parameter K called the curvature of spacetime.

Flat Spacetime if $K = 0$ spacetime is flat, it could be either infinite or finite.

Hyperspherical Spacetime if $K > 0$ spacetime is spherical and the universe must be finite.

Hyperbolic Spacetime if $K < 0$ spacetime looks like a saddle, and it could be either infinite or finite.

If we choose spherical coordinates

$$x^\mu = (t, r, \theta, \varphi) \tag{20.18}$$

then the metric is

$$\mathrm{d}s^2 = c^2\, \mathrm{d}t^2 + a^2(t)\left[\frac{\mathrm{d}r^2}{1 - Kr^2} + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\varphi^2\right)\right] \tag{20.19}$$

in terms of unit analysis, we can have two alternative options. The first is

$$[a] = 1 \quad \& \quad [r] = L \implies [K] = L^{-2} \tag{20.20}$$

in which case you can normalize a such that $a(t_0) = 1$. The other is

$$[a] = L \quad \& \quad [r] = 1 \implies [K] = 1 \tag{20.21}$$

in which case you can normalize K such that $K = 0, \pm 1$. We can define a typical curvature radius:

$$R_{curv} = \frac{a(t)}{\sqrt{|K|}} \tag{20.22}$$

Lecture 21.

Thu, January 15th, 2026

A note on notation for the scale factor: there are several notations used. One defines the scale factor as $a(t)$ as we have done, but another defines it as $R(t)$. Both notations are equivalent. In the notation $a(t)$, the scale factor is often normalized such that $a(t_0) = 1$ where t_0 is the current time. In the notation $R(t)$, the scale factor is often normalized such that $R(t_0) = R_0$ where R_0 is the current scale factor, which has units of length.

Using the $R(t)$ notation we can write the metric, known as the Friedmann-Lemaître-Robertson-Walker metric

$$\mathrm{d}s^2 = (c\,\mathrm{d}t)^2 - \mathrm{d}l^2 \tag{21.1}$$

$$\mathrm{d}l^2 = R^2(t)\left[\frac{\mathrm{d}r^2}{1-Kr^2} + r^2\left(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2\right)\right] \tag{21.2}$$

The physical distance to a galaxy located at r is given by

$$l = \int \mathrm{d}l = R(t)\int_0^r \frac{\mathrm{d}r'}{\sqrt{1-Kr'^2}} = \begin{cases} R(t)\arcsin r & K = 1 \\ R(t)r & K = 0 \\ R(t)\,\mathrm{arcsinh}\,r & K = -1 \end{cases} \tag{21.3}$$

and the velocity is

$$v = \frac{\mathrm{d}l}{\mathrm{d}t} = \dot{R}(t)\int \frac{\mathrm{d}r'}{\sqrt{1-Kr'^2}} = \frac{\dot{R}}{R}\cdot R\int \frac{\mathrm{d}r'}{\sqrt{1-Kr'^2}} = H(t)l \tag{21.4}$$

which is Hubble's Law again.

7.5 What Is The Scale Factor?

The Einstein equation is

$$G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} \tag{21.5}$$

The metric is given by the FLRW metric:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\frac{R^2(t)}{1-Kr^2} & 0 & 0 \\ 0 & 0 & -R^2(t)r^2 & 0 \\ 0 & 0 & 0 & -R^2(t)r^2\sin^2\theta \end{pmatrix} \tag{21.6}$$

To get to the Einstein tensor $G_{\mu\nu}$ there is a simple formula:

$$G_{\mu\nu} = \mathfrak{R}_{\mu\nu} - \frac{1}{2}\mathfrak{R}g_{\mu\nu} \tag{21.7}$$

where

$$\mathfrak{R} = g^{\beta\gamma}\mathfrak{R}_{\beta\gamma} \tag{21.8}$$

$$\mathfrak{R}_{\beta\gamma} = g^{\delta}_{\alpha}\mathfrak{R}^{\alpha}_{\beta\delta\gamma} \tag{21.9}$$

$$\mathfrak{R}^{\alpha}_{\beta\delta\gamma} = \partial_\gamma\Gamma^{\alpha}_{\beta\delta} - \partial_\delta\Gamma^{\alpha}_{\beta\gamma} + \Gamma^{\alpha}_{\rho\gamma}\Gamma^{\rho}_{\beta\delta} - \Gamma^{\alpha}_{\rho\delta}\Gamma^{\rho}_{\beta\gamma} \tag{21.10}$$

$$\Gamma^{\mu}_{\sigma\nu} = \frac{1}{2}g^{\mu\nu}\left(\partial_\nu g_{\sigma\rho} + \partial_\sigma g_{\nu\beta} - \partial_\rho g_{\sigma\nu}\right) \tag{21.11}$$

The first line is called the Ricci scalar, the second line is called the Ricci tensor, the third line is called the Riemann curvature tensor, and the last line is called the Christoffel symbols. Skipping lots of algebra we get

$$G_{00} = \frac{3}{c^2R^2}\left(\dot{R}^2 + Kc^2\right) \tag{21.12}$$

$$G_{11} = -\frac{R\ddot{R} + \dot{R}^2 + Kc^2}{c^2\left(1 - Kr^2\right)} \tag{21.13}$$

The elements off the diagonal are zero. G_{22} and G_{33} can be found from symmetry.

7.6 The Momentum Energy Tensor

The elements of the momentum energy tensor are

$$T_{00} = \text{Energy Density} \tag{21.14}$$

$$T_{0i} = \text{Momentum Flux} \tag{21.15}$$

$$T_{ii} = \text{Isotropic Pressure} \tag{21.16}$$

$$T_{ij} = \text{Non-Isotropic Pressure From Shear Stress} \tag{21.17}$$

In a homogenous and isotropic universe $T_{\mu\nu}$ is diagonal and given by

$$T_{\mu\mu} = \left(P + \rho c^2\right)\frac{v_\mu v\mu}{c^2} - Pg_{\mu\mu} \tag{21.18}$$

and the four-velocity is

$$v^\mu = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \gamma\left(\frac{c}{u}\right) \tag{21.19}$$

We now have

$$T_{00} = \rho c^2 \tag{21.20}$$

$$T_{11} = \frac{PR^2(t)}{1 - Kr^2} \tag{21.21}$$

comparing with G_{00} and G_{11} we get the Friedmann equations:

$$\frac{\dot{R}^2 + Kc^2}{R^2} = \frac{8\pi}{3}G\rho \tag{21.22}$$

$$\frac{2\ddot{R}}{R} + \frac{\dot{R}^2 + KC^2}{R^2} = -\frac{8\pi}{c^2}GP \tag{21.23}$$

The first equation describes how the expansion rate depends on the energy density and curvature, and the second equation describes how the acceleration of the expansion depends on the energy density and pressure.

Lecture 22.

Mon, January 19th, 2026

We can rearrange the Friedman equations into

$$\left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi}{3}G\rho - \frac{Kc^2}{R^2} \tag{22.1}$$

$$\frac{\ddot{R}}{R} = -\frac{4\pi G}{3c^2}\left(\rho c^2 + 3P\right) \tag{22.2}$$

We'll investigate these equations: in a a static universe $\dot{R} = 0$ and $\ddot{R} = 0$ and if we plug that into the 2nd equation, where pressure and density must be non-negative, the only solution is an empty universe with no density and no pressure. By the way, this troubled Einstein so much that he added a cosmological constant to his equations to allow for a static universe. We'll add it later. Leaving this scenario behind, the 2nd equation says that for a non-empty universe the expansion must be decelerating: $\ddot{R} < 0$. Observations show that the expansion is actually accelerating, so something must be missing from the equation. We're looking at the incomplete equations for pedagogical purposes only. Sometimes it's useful to combine the equations into what's called the third Friedmann equation or the fluid equation:

$$\dot{\rho}c^2 = -3\frac{\dot{R}}{R}\left(\rho c^2 + P\right) \tag{22.3}$$

this equation can also be reached using thermodynamics. Our goal now is to find an expression for $R(t)$. We'll look at different ages of the universe. In the matter dominated era, radiation pressure is negligible compared to energy density, so $P \ll \rho c^2$. The fluid equation becomes

$$\frac{\dot{\rho}}{\rho} = -3\frac{\dot{R}}{R} \tag{22.4}$$

which is a simple differential equation with solution

$$\rho(t) \propto R(t)^{-3} \tag{22.5}$$

in the radiation dominated era, we'll replace the element ρc^2 with u_{rad} and earlier in the course we found that $P = u_{rad}/3$. The fluid equation becomes

$$\dot{u}_{rad} = -3\frac{\dot{R}}{R}\left(u_{rad} + \frac{1}{3}u_{rad}\right) = -4\frac{\dot{R}}{R}u_{rad} \tag{22.6}$$

with solution

$$u_{rad}(t) \propto R(t)^{-4} \tag{22.7}$$

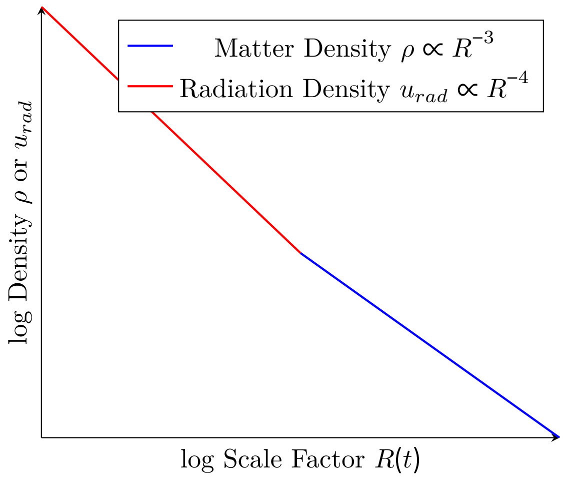


Figure 22.1: Density as a function of scale factor, showing the difference between matter and radiation domination. It can be shown that $a = \frac{R}{R_0} = \frac{1}{3500}$ and $\frac{t}{t_0} = \frac{1}{2\times10^5\ \text{unit}}$.

we can then find that in the radiation dominated era, $R \propto t^{1/2}$ and in the matter dominated era, $R \propto t^{2/3}$. In late times the element $K\frac{c^2}{r^2}$ becomes important, this is called the curvature dominated era. We'll look at the three possibilities for K. If $K = 0$ (the flat universe) then from the first Friedmann equation we have

$$\frac{\dot{R}}{R} = H^2 = \sqrt{\frac{8\pi G\rho}{3}} \tag{22.8}$$

from observations we know $H_0 = \frac{8\pi G}{3}\rho_0$. We define a critical density as

$$\rho_c = \frac{3H^2}{8\pi G} \tag{22.9}$$

inserting current values we find $\rho_c = 1.4 \times 10^{11} \frac{M_\odot}{\mathrm{Mpc}^3}$. We can then define a density parameter as

$$\Omega = \frac{\rho}{\rho_c} \tag{22.10}$$

if $\Omega = 1$ the universe is flat, if $\Omega > 1$ the universe is closed, and if $\Omega < 1$ the universe is open. Continuing to the case $K = 1$, the closed universe, we have

$$\left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi G}{3}\rho - \frac{c^2}{R^2} \tag{22.11}$$

The two elements on the right hand side will eventually balance out, at which point $\dot{R} = 0$ and the universe stops expanding and starts contracting. The maximum radius can be found by setting the right hand side to zero:

$$R_{max} = \sqrt{\frac{3c^2}{8\pi G\rho}} \tag{22.12}$$

In the case $K = -1$, the open universe,

$$\left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi G}{3}\rho + \frac{c^2}{R^2} \tag{22.13}$$

and in large times $\dot{R} \to c$. Overall if we graph $R(t)$ for the three cases we get something like this: Another way of writing the Friedmann equations is by defining $\Omega_{m,o} = \frac{\rho_{m,0}}{\rho_{c,0}}$, $\Omega_{rad,0} = \frac{\rho_{rad,0}}{\rho_{c,0}}$, $\Omega_0 = \sum \Omega_{i,0}$, $a = \frac{R}{R_0}$, and $H = \frac{\dot{R}}{R}$. The first Friedmann equation becomes

$$\left(\frac{H}{H_0}\right)^2 = \frac{\Omega_{m,0}}{a^3} + \frac{\Omega_{rad,0}}{a^4} + \frac{1-\Omega_0}{a^2} \tag{22.14}$$

In the case of a flat universe, $\Omega_0 = 1$ and the last term vanishes. If $K = 1$ then $\Omega_0 > 1$ and if $K = -1$ then $\Omega_0 < 1$. In the general case we can find $a(t)$ from this equation using $H = \frac{\dot{a}}{a}$ and solving the differential equation. For example, let's find the age of the universe today by solving for a flat universe. We'll ignore the radiation term since it's negligible today. We're left with

$$\left(\frac{\dot{a}}{a}\right)^2 = H_0^2\frac{\Omega_{m,0}}{a^3} \tag{22.15}$$

Since $\Omega_{m,0}$ is the only contributor to Ω_0 in this scenario, it must be equal to 1. Rearranging and integrating we get

$$\int_0^{t_0} H_0\,\mathrm{d}t = H_0 t_0 = \int_0^1 \left(\frac{1}{a^3}\right)^{-1/2} \mathrm{d}a = \frac{2}{3} \tag{22.16}$$

therefore

$$t_0 = \frac{2}{3H_0} \tag{22.17}$$

which is about 9 Gyr. This is lower than the actual age of the universe, about 13.8 Gyr, because we ignored dark energy which is causing the expansion to accelerate. Back to Einstein's issue with the friedmann equations predicting a non-static universe, he added a cosmological constant Λ to the equations:

$$\left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi G}{3}\rho - \frac{Kc^2}{R^2} + \frac{1}{3}\Lambda \tag{22.18}$$

$$\frac{\ddot{R}}{R} = -\frac{4\pi G}{3c^2}\left(\rho c^2 + 3P\right) + \frac{1}{3}\Lambda \tag{22.19}$$

$$\dot{\rho}c^2 = -3\frac{\dot{R}}{R}\left(\rho c^2 + P\right) \tag{22.20}$$

Later observations by hubble showed that the universe is expanding, so Einstein called the addition of the cosmological constant his 'biggest mistake'. Recent observations however show that the expansion is accelerating, so the cosmological constant (or something like it) is used again, with the interpretation that it represents dark energy. We can see that if Λ is large enough, the 2nd equation can give $\ddot{R} > 0$. If we define the dark energy density as

$$\rho_\Lambda = \frac{\Lambda}{8\pi G} \tag{22.21}$$

and the dark energy density as

$$\epsilon_\Lambda = \rho_\Lambda c^2 = \frac{\Lambda c^2}{8\pi G} \tag{22.22}$$

and

$$\Omega_\Lambda = \frac{\rho_\Lambda}{\rho_c} = \frac{\Lambda}{3H^2} \tag{22.23}$$

we can rewrite the first Friedmann equation as

$$\left(\frac{H}{H_0}\right)^2 = \frac{\Omega_{m,0}}{a^3} + \frac{\Omega_{rad,0}}{a^4} + \Omega_\Lambda + \frac{1-\Omega_0}{a^2} \tag{22.24}$$

We can see that since Λ is constant, so is the energy density ϵ_Λ. Let's look into this oddity. The state equation is

$$\dot{\epsilon} = -3\frac{\dot{R}}{R}(\epsilon + P) \tag{22.25}$$

if we define $\omega = \frac{P}{\epsilon}$ then

$$\frac{\dot{\epsilon}}{\epsilon} = -3\frac{\dot{R}}{R}(1+\omega) \tag{22.26}$$

for non-relativistic matter $P \ll \rho c^2$ which gives $\omega \simeq 0$. For relativistic matter or radiation we get $\omega = 1/3$. For dark energy we want $\dot{\epsilon} = 0$ which gives $\omega = -1$. A theory that was proposed for the constant energy density is that of vacuum energy. In quantum field theory the vacuum has an energy associated with it, and this energy density is constant as space expands. This is given by

$$\epsilon_{vac} = \frac{E_P}{l_P^3} = 3 \times 10^{127}\,\mathrm{eV\,cm^{-3}} \tag{22.27}$$

but observations give

$$\epsilon_\Lambda \approx 4000\,\mathrm{eV\,cm^{-3}} \tag{22.28}$$

which is a huge discrepancy. This is called the cosmological constant problem and is an open question in physics. Expansion with the cosmological constant becomes dominant at late times, when the matter and radiation densities have dropped enough.

$$H^2 = \left(\frac{\dot{R}}{R}\right)^2 = \frac{1}{3}\Lambda \tag{22.29}$$

which has solution

$$R(t) \propto e^{Ht} = e^{\sqrt{\frac{\Lambda}{3}}t} \tag{22.30}$$

Lecture 23.

Thu, January 22nd, 2026

There are two kinds of horizons: the particle horizon and the cosmological horizon. The particle horizon is the maximum distance from which light could have reached us since the beginning of the universe. For a particle dominated universe, the particle horizon is given by

$$x_{ph} = 3ct_0 \tag{23.1}$$

and for a Λ dominated universe

$$x_{ph} = \frac{c}{H_0} = \frac{c}{\sqrt{1/3}} = 5.5\,\text{Gpc} \tag{23.2}$$

The cosmological horizon is the maximum distance light we emit now can travel to in the future. To calculate it, we look back to the FHW metric:

$$ds^2 = c^2 dt^2 - R(t)^2 \left[\frac{\mathrm{d}r^2}{1 - Kr^2} + r^2\,\mathrm{d}\Omega^2 \right] \tag{23.3}$$

we'll assume we're looking at light so $ds^2 = 0$ and only in the radial direction so $\mathrm{d}\Omega = 0$. This gives us

$$c\,\mathrm{d}t = R(t)\frac{\mathrm{d}r}{\sqrt{1 - Kr^2}} \tag{23.4}$$

For convenience, we'll assume a flat universe so $K = 0$ and we have

$$c\,\mathrm{d}t = R(t)\,\mathrm{d}r \tag{23.5}$$

then to move to the comoving coordinates we use $a = R/R_0$

$$c\,\mathrm{d}t = R_0 a\,\mathrm{d}r \tag{23.6}$$

then we integrate

$$x_{ch} = \int_0^{x_{ch}} \mathrm{d}x = \int_{t_0}^{\infty} \frac{c\,\mathrm{d}t}{a(t)} \tag{23.7}$$

In a matter dominated universe, $a(t) = (t/t_0)^{2/3}$ so

$$x_{ch} = ct_0 \int_1^{\infty} u^{-2/3}\,\mathrm{d}u \tag{23.8}$$

which diverges, so in a matter dominated universe there is no cosmological horizon. In a Λ dominated universe, $a(t) = e^{H_0 t}$ so

$$x_{ch} = c\int_{t_0}^{\infty} e^{-Ht}\,\mathrm{d}t = \frac{c}{H} e^{-Ht_0} \tag{23.9}$$

translating to physical distance

$$\mathrm{d}p = \int \mathrm{d}l = R\int \frac{\mathrm{d}r}{\sqrt{1 - Kr^2}} \tag{23.10}$$

for $K = 0$ this is just

$$dp = Rr = \frac{R}{R_0} R_0 r = a(t)x \quad (23.11)$$

so the physical distance to the cosmological horizon is

$$dp|_{t=t_0} = \frac{c}{H} e^{-Ht_0} e^{Ht_0} = \frac{c}{H} \quad (23.12)$$

	Comoving (x)	Physical (dp)
Galaxy i	$x_i = c_i$ constant	$c_i e^{Ht}$
Particle horizon	$x_{ph} = \frac{c}{H}$	$dp_{ph} = \frac{c}{H} e^{Ht}$
Cosmological horizon	$x_{ch} = \frac{c}{H} e^{-Ht_0}$	$dp_{ch} = \frac{c}{H}$

Table 23.1: Comparison of comoving and physical distances

7.7 Observations

The first and most important observation is of redshifted light. We can find a connection between the redshift and the scale factor. The density of energy of radiation is given by

$$\epsilon_{rad} \propto R^{-4} \tag{23.13}$$

but on the other hand, the energy density is also given by

$$\epsilon_{rad} = \frac{E_{rad}}{R^3} = \frac{hc}{\lambda R^3} \tag{23.14}$$

but the wavelength is proportional to R so

$$\lambda_{obs} = \frac{1}{a} \lambda_{em} \tag{23.15}$$

and in terms of redshift

$$1 + z = \frac{1}{a} \tag{23.16}$$

We'll find another expression for distance to proceed. We know that

$$dp = \int dl = R(t) \int \frac{dr}{\sqrt{1 - Kr^2}} \tag{23.17}$$

and at time t_0 this is

$$dp = R_0 \int \frac{dr}{\sqrt{1 - Kr^2}} \tag{23.18}$$

on the other hand, looking at a photon emitted at time t_{em} and observed at time t_0, using the metric we have

$$c\,dt = R(t) \frac{dr}{\sqrt{1 - Kr^2}} = R_0 a(t) \frac{dr}{\sqrt{1 - Kr^2}} \tag{23.19}$$

rearranging and integrating gives

$$\int_{t_{em}}^{t_0} \frac{c\,dt}{a(t)} = R_0 \int_0^r \frac{dr}{\sqrt{1 - Kr^2}} \tag{23.20}$$

so

$$dp = \int_{t_{em}}^{t_0} \frac{c\,dt}{a(t)} \tag{23.21}$$

Now that we have this,

$$dp = \int \frac{c\,dt}{a} = \int \frac{c \frac{dt}{da}\,da}{a} \tag{23.22}$$

$$= \int \frac{c\,da}{\dot{a} a} = c \int \frac{da}{\left(\frac{\dot{a}}{a}\right) a^2} \tag{23.23}$$

inserting the Friedmann equation

$$\mathrm{d}p = \frac{c}{H_0} \int \frac{\mathrm{d}a}{\sqrt{\frac{\Omega_m}{a^3} + \frac{\Omega_{rad}}{a^4} + \Omega_\Lambda + \frac{1-\Omega}{a^2}} a^2} \tag{23.24}$$

Changing variables to redshift using $a = \frac{1}{1+z}$

$$\mathrm{d}p = \frac{c}{H_0} \int_0^{z_{em}} \frac{\mathrm{d}z}{\sqrt{\Omega_m(1+z)^3 + \Omega_{rad}(1+z)^4 + \Omega_\Lambda + (1-\Omega)(1+z)^2}} \tag{23.25}$$

Lecture 24.

Mon, January 26th, 2026

7.8 Distance Measurements

To measure distances in cosmology there are two main methods: standard candles and standard rulers. Standard candles are mostly type Ia supernovae, and the standard ruler is usually the cosmic microwave background (CMB).

Standard Candles

To use a standard candle we use the relationship between luminosity and flux:

$$F = \frac{L}{4\pi d^2} \tag{24.1}$$

and since we know L and we can measure F, we can find d. The problem is that this is actually a lie because this relationship doesn't hold in an expanding universe. The actual relationship comes from

$$E_{obs} = \frac{E_{em}}{\frac{R_0}{R}} = aE_{em} = \frac{E_{em}}{1+z} \tag{24.2}$$

We also need to take into account cosmological time dilation. Using the metric

$$\mathrm{d}s^2 = c^2\,\mathrm{d}t^2 - R^2(t)\left(\frac{\mathrm{d}r^2}{1-Kr^2} + r^2\,\mathrm{d}\Omega^2\right) \tag{24.3}$$

and for photons on the radial direction $\mathrm{d}s = 0$ and $\mathrm{d}\Omega = 0$, so

$$\int \frac{c\,\mathrm{d}t}{R(t)} = \int_0^{r_e} \frac{\mathrm{d}r}{\sqrt{1-Kr^2}} \tag{24.4}$$

If we look at two photons emitted at t_{em} and $t_{em} + \Delta t_{em}$ and observed at t_0 and $t_0 + \Delta t_0$, we have for the first photon

$$\int_{t_{em}}^{t_0} \frac{c\,\mathrm{d}t}{R(t)} = \int_0^{r_e} \frac{\mathrm{d}r}{\sqrt{1-Kr^2}} \tag{24.5}$$

and for the second photon

$$\int_{t_{em}+\Delta t_{em}}^{t_0+\Delta t_0} \frac{c\,\mathrm{d}t}{R(t)} = \int_0^{r_e} \frac{\mathrm{d}r}{\sqrt{1-Kr^2}} \tag{24.6}$$

the right hand sides of both equations are the same which gives us

$$c\int_{t_{em}}^{t_0} \frac{\mathrm{d}t}{R(t)} = c\int_{t_{em}+\Delta t_{em}}^{t_0+\Delta t_0} \frac{\mathrm{d}t}{R(t)} \tag{24.7}$$

skipping some algebra we get

$$\frac{\Delta t_0}{\Delta t_{em}} = \frac{R_0}{R_{em}} = \frac{1}{a} = 1+z \tag{24.8}$$

So the observed luminosity is

$$L_{obs} = \left.\frac{\mathrm{d}E}{\mathrm{d}t}\right|_{obs} = \frac{E_{obs}}{\Delta t_{obs}} = \frac{E_{em}/(1+z)}{\Delta t_{em}(1+z)} = \frac{L_{em}}{(1+z)^2} \tag{24.9}$$

So the flux is

$$F = \frac{L_{obs}}{4\pi d^2} = \frac{L_{em}}{4\pi d^2 (1+z)^2} \tag{24.10}$$

Sometimes we define the luminosity distance d_L as

$$F = \frac{L_{em}}{4\pi d_L^2} \tag{24.11}$$

We can use this for supernovae meaasurements, since they have a known luminosity, and we can measure their redshift, so we can find the luminosity distance d_L. Looking at measurements of type Ia supernovae and their fits to different cosmological models, we find that the data fits a model with $\Omega_m \approx 0.3$ and $\Omega_\Lambda \approx 0.7$, which is evidence for dark energy. If we look at the $\Omega_\Lambda - \Omega_m$ phase space, we're able to place constraints on the parameters, as shown in figure .

Standard Ruler

In a static universe if we have a ruler of length l at a distance d perpendicular to our line of sight, and it takes an angle $\delta\theta$, we have

$$d = \frac{l}{\delta\theta} \tag{24.12}$$

for an expanding universe we can use the metric again.

$$\mathrm{d}s^2 = -c^2 \,\mathrm{d}t^2 + R^2(t)\left(\frac{\mathrm{d}r^2}{1-Kr^2} + r^2 \,\mathrm{d}\theta^2 + \dots\right) \tag{24.13}$$

If we look at the two sides of the ruler as two events of photon emission, A and B, which happened at the same time, we have $\mathrm{d}t = 0$ so

$$\mathrm{d}s^2 = R^2(t)\left(\frac{\mathrm{d}r^2}{1-Kr^2} + r^2 \,\mathrm{d}\theta^2\right) \tag{24.14}$$

If the ruler is perpendicular to our line of sight, $\mathrm{d}r = 0$ so

$$\mathrm{d}s^2 = R^2(t) r^2 \,\mathrm{d}\theta^2 \tag{24.15}$$

so the physical length of the ruler is

$$l = R(t) r \,\mathrm{d}\theta = Rar \,\mathrm{d}\theta \tag{24.16}$$

a result we had earlier for $\mathrm{d}p$ was

$$\mathrm{d}p = R_0 \int \frac{\mathrm{d}r}{\sqrt{1-Kr^2}} = R_0 r \tag{24.17}$$

which is specifically for the case $K = 0$ which we'll focus on. So we can write

$$l = a \,\mathrm{d}p \,\mathrm{d}\theta \tag{24.18}$$

or in terms of redshift

$$l = \frac{\mathrm{d}p}{1+z} \,\mathrm{d}\theta \tag{24.19}$$

if we define angular distance d_A as

$$d_A = \frac{\mathrm{d}p_0}{1+z} \tag{24.20}$$

we have

$$l = d_A \,\mathrm{d}\theta \tag{24.21}$$

7.9 Cosmic Microwave Background (CMB)

In 1949 it was predicted by Alper and Herman that if the universe started in a hot dense state, there should be a background of black body radiation left over from that time which cools with time. We know that

$$\epsilon_{rad} \propto a^{-4} \tag{24.22}$$

and the wavelength of the radiation scales as

$$\lambda \propto 1 + z \tag{24.23}$$

on the other hand

$$\epsilon_{rad} \propto T^4 \tag{24.24}$$

so

$$T \propto \frac{T_0}{a} = T_0(1+z) \tag{24.25}$$

from this analysis it was predicted that the temperature of the CMB today should be around 5 K. In 1941 McKellar measured the temperature of interstellar CN molecules and found a floor temperature of around 2.3 K. In 1965 Penzias and Wilson conducted an experiment to measure the noise in a radio antenna and found an excess noise (about 100 times more than they expected) at all times of the day coming from all directions which they couldn't explain. After trying to locate the source of the noise they ended up concluding that it was actually the CMB, and they got the Nobel prize for it in 1978. The temperature they measured was around 3 K.

Source Of The CMB

If we look at the early universe, there was very hot gas and radiation, it was ionized, and there were many interactions. From the process of thermalisation, the radiation became a black body spectrum.

$$B_\nu = \frac{2h\nu^3}{c^2}\frac{\mathrm{d}\nu}{e^{\frac{h\nu}{kT}} - 1} \tag{24.26}$$

$$\epsilon_{rad} = a_{rad}T^4 \tag{24.27}$$

$$T(a) = \frac{T_0}{a} = T_0(1+z) \tag{24.28}$$

What are the recombination temperature and time? Naively we can look at the ionization energy of hydrogen which is 13.6 eV which corresponds to a temperature of around 1×10^6 K. This is incorrect, and to find the correct temperature we can solve

$$\langle\sigma\nu\rangle\, n_H n_\gamma = n_H + n_e\alpha(t) \tag{24.29}$$

which is the ionization rate being equal to the recombination rate, and if we solve it we find that the recombination temperature is around 3000 K. This corresponds to a redshift of around $z \approx 1100$. To find the time at which this happened we can use the Friedmann equation (simplified for a matter dominated flat universe)

$$a = \left(\frac{t}{t_0}\right)^{2/3} \tag{24.30}$$

so

$$t = \frac{t_0}{(1+z)^{3/2}} \approx 370\,000\,\mathrm{yr} \tag{24.31}$$

So the CMB was emitted around 370 000 yr after the Big Bang at a temperature of around 3000 K. The surface from which the CMB was emitted is called the last scattering surface. The distance to it is approximately the same as that of the particle horizon. The accurate calculation for it gives around $3.4ct_0 = 46.6$ Gly. The comoving coordinate of the last scattering surface is

$$x_{lss} = R_0 r|_{lss} = 3.4ct_0 \tag{24.32}$$

Back to measurements of the CMB, the first one used the horn antenna, but later on satellites were used, such as WMAP and Planck. These satellites measured the intensity of the CMB across difference frequiencies and found that it matches a black body spectrum with a temperature of around 2.725 K very accurately. In fact the CMB is the most perfect black body spectrum ever measured. This average is across the entire sky but if we look at smaller scales we find anisotropies in the CMB. The strongest anisotropy is a dipole anisotropy which is due to the motion of the earth with respect to the CMB rest frame. After removing this dipole anisotropy, we find the radiation is very isotropic with fluctuations $\frac{\delta T}{\langle T \rangle} \approx 10^{-5}$.

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